

Medical Image Analysis

(Multi)Atlas-Patch based Segmentation

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- Atlas Based segmentation
- Multi-Atlas based segmentation
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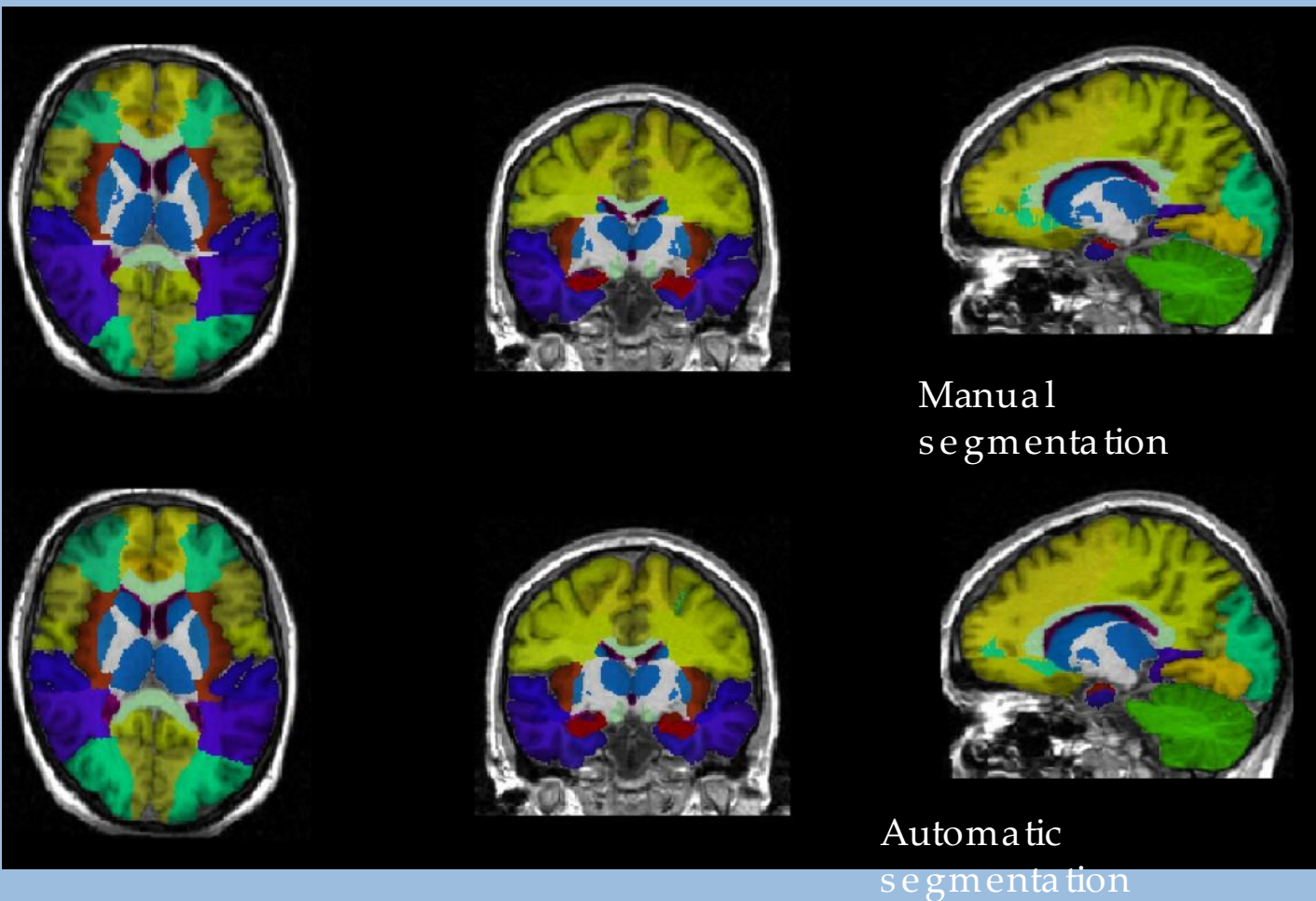
With slide credits

D. Rueckert

L. Wang

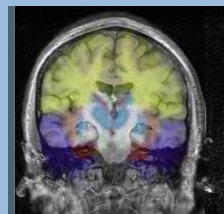
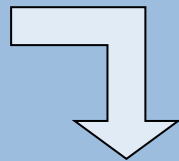
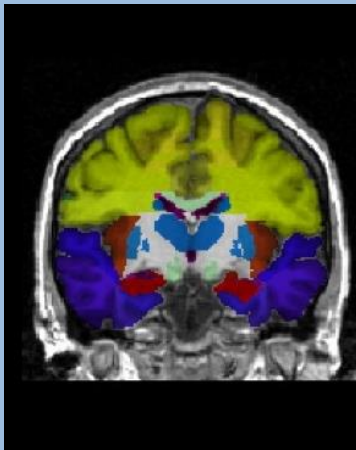
T. Tong

Segmentation using registration

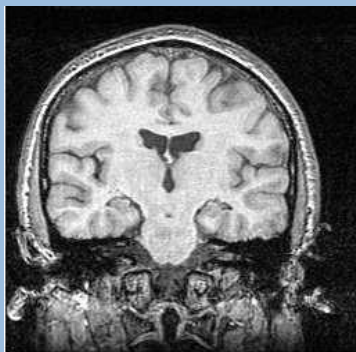


Segmentation via Registration

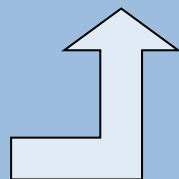
ATLAS



Registration
Atlas-to-image



Patient Image

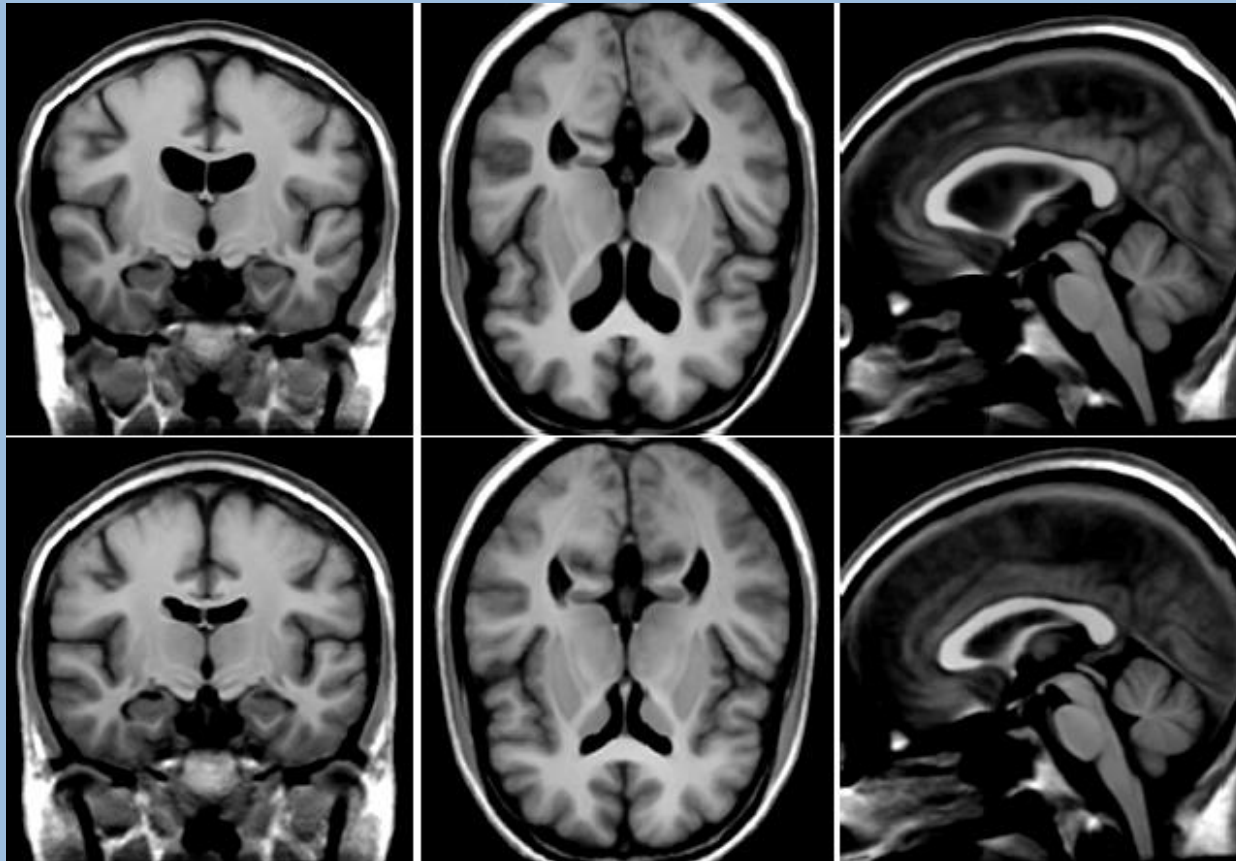


Factors influencing the quality of the result:

- Image similarity
- Presence of pathology
- Quality of the registration
- Likelihood of patient image

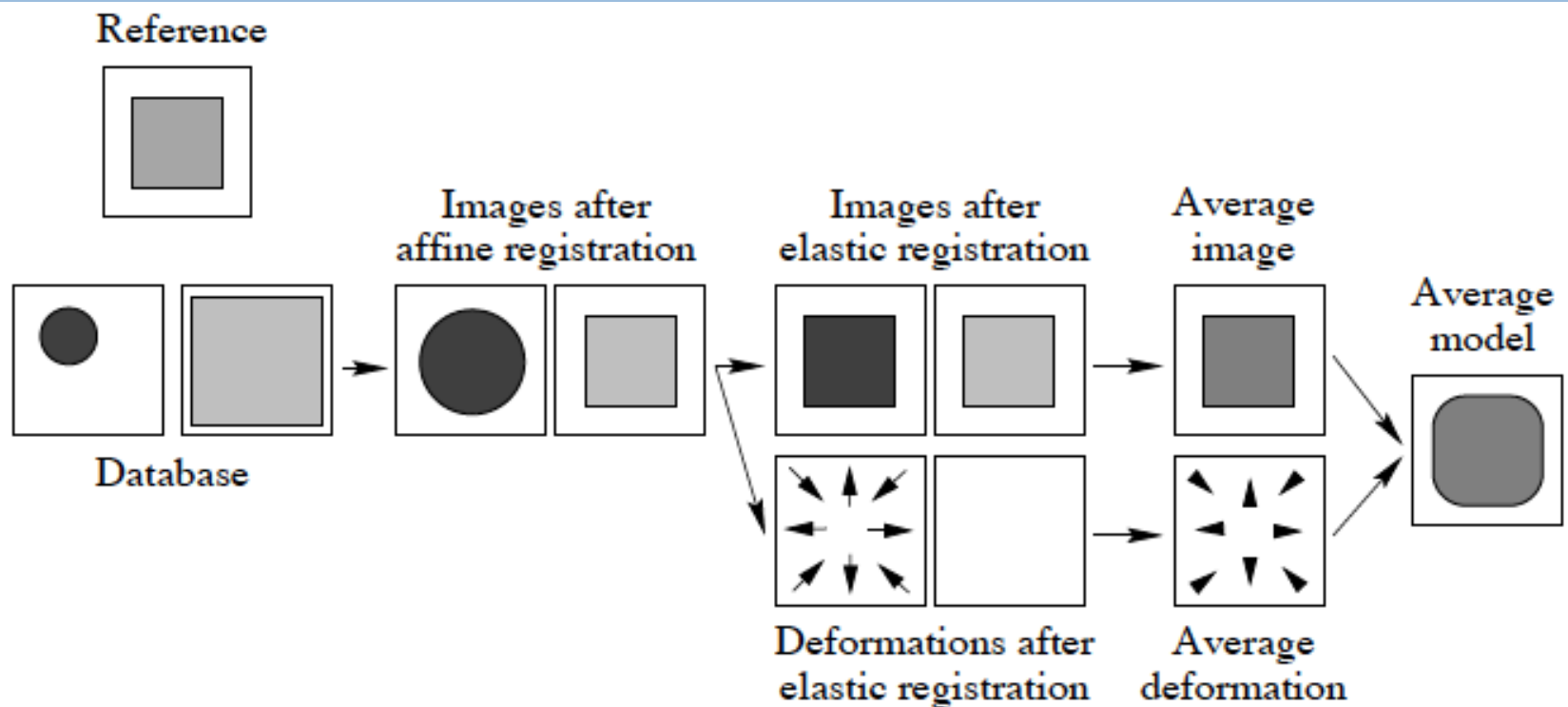
Bias in reference selection and natural coordinates reference

Not the same! ☹



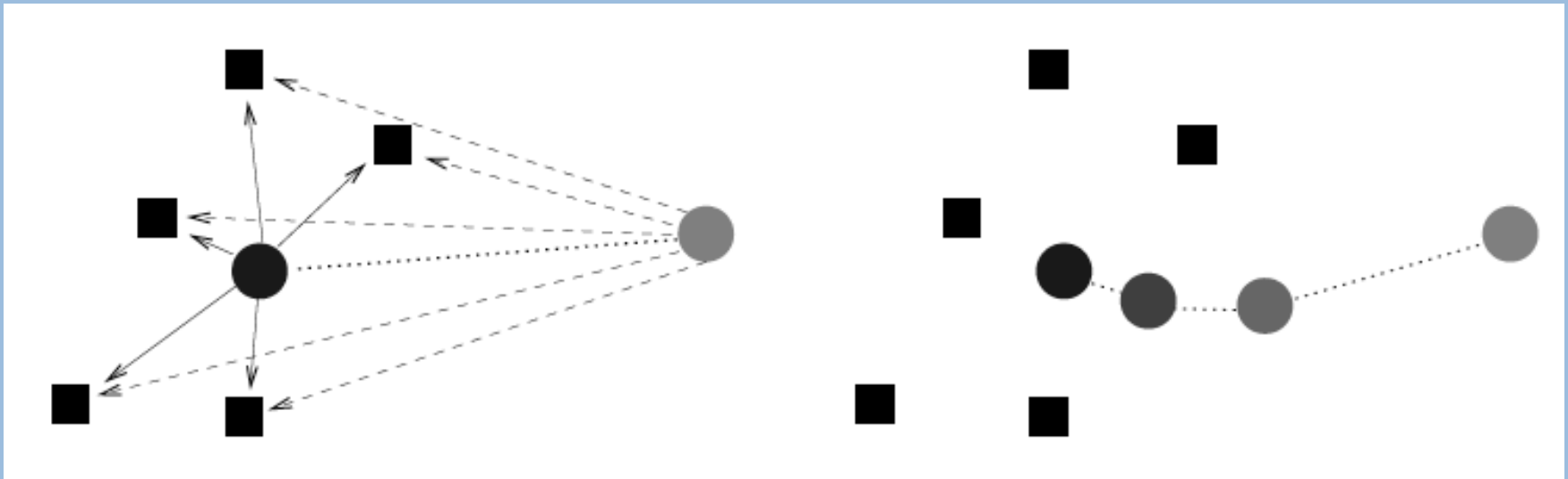
Bias in reference selection and natural coordinates reference

Computing mean image: Guimond et al. CVIU 2000



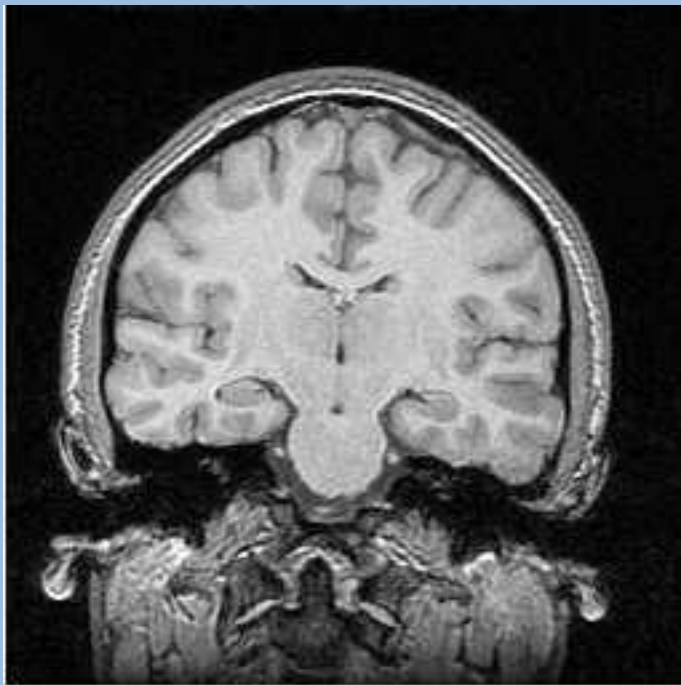
Bias in reference selection and group-wise modelling

Computing mean image: Guimond et al. Miccai 2000

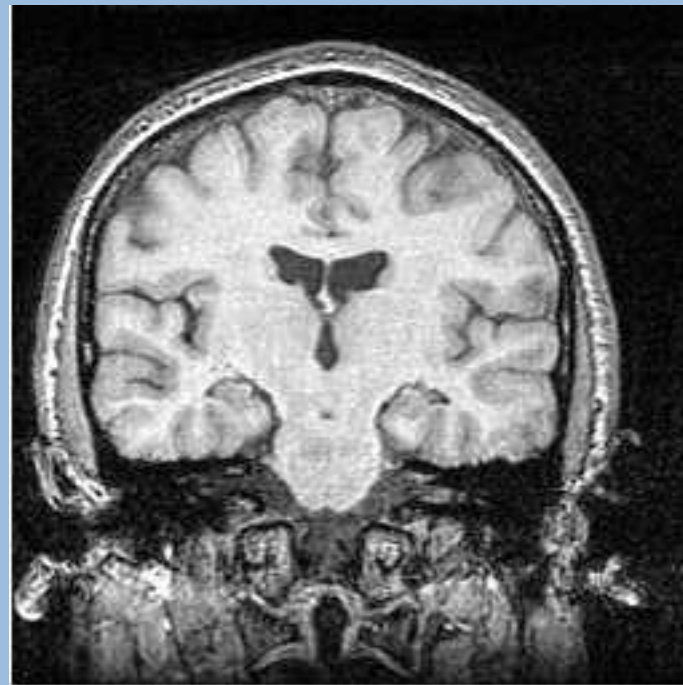


Bias in reference selection and group-wise modelling

Computing mean image: Guimond et al. Miccai 2000



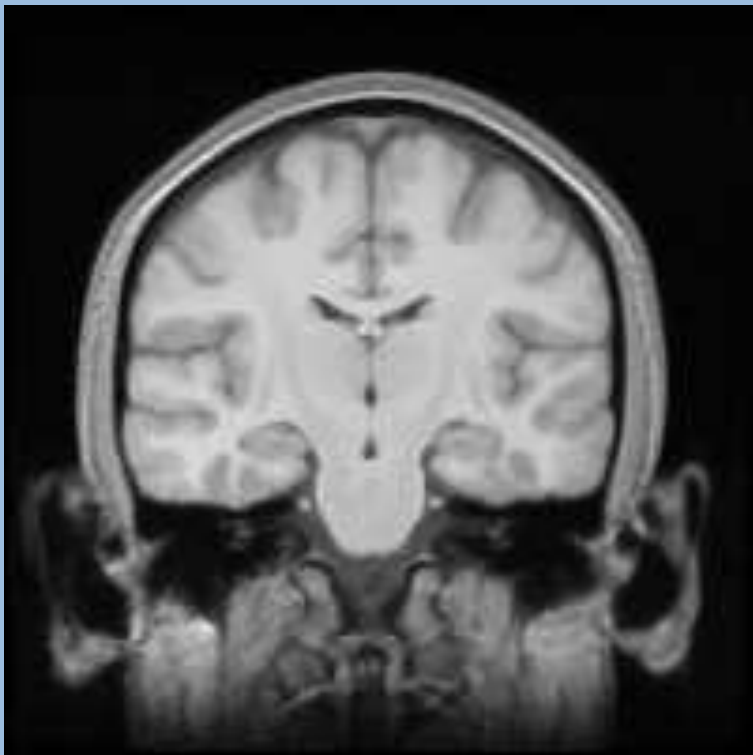
Reference 1



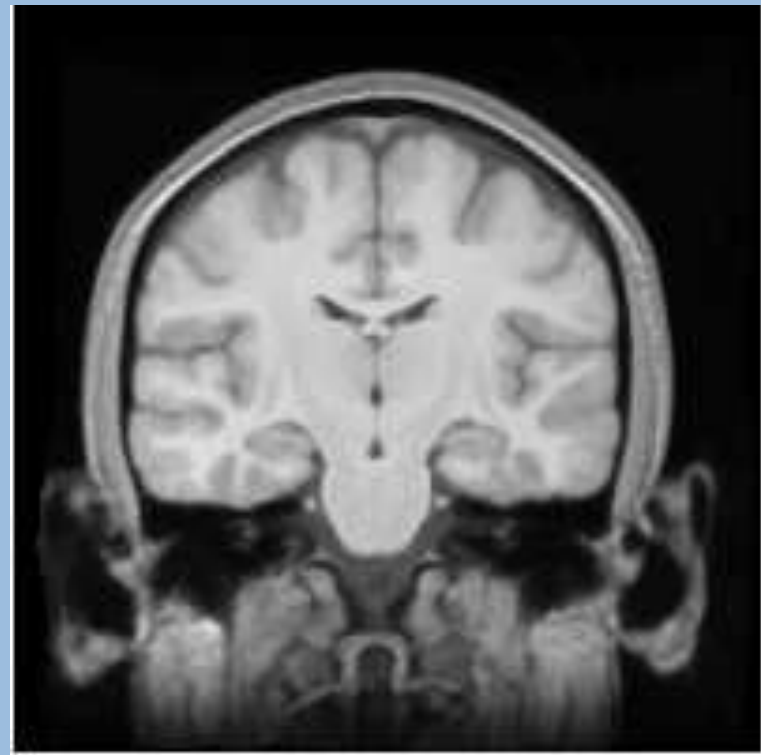
Reference 2

Bias in reference selection and group-wise modelling

Computing mean image: Guimond et al. Miccai 2000

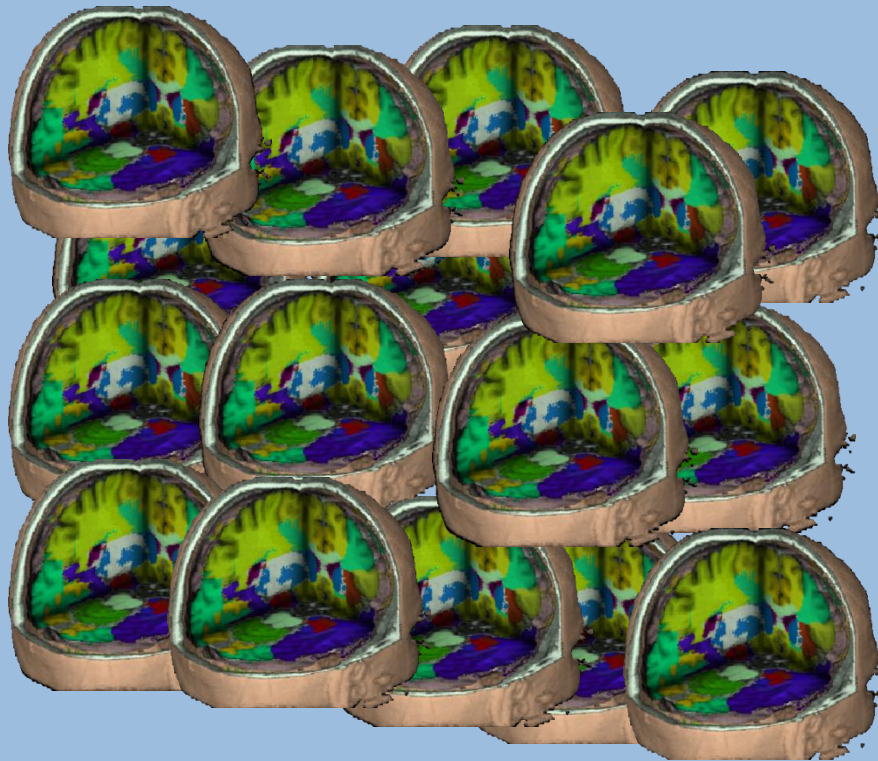


Average model using ref 1



Average Intensity using ref 2

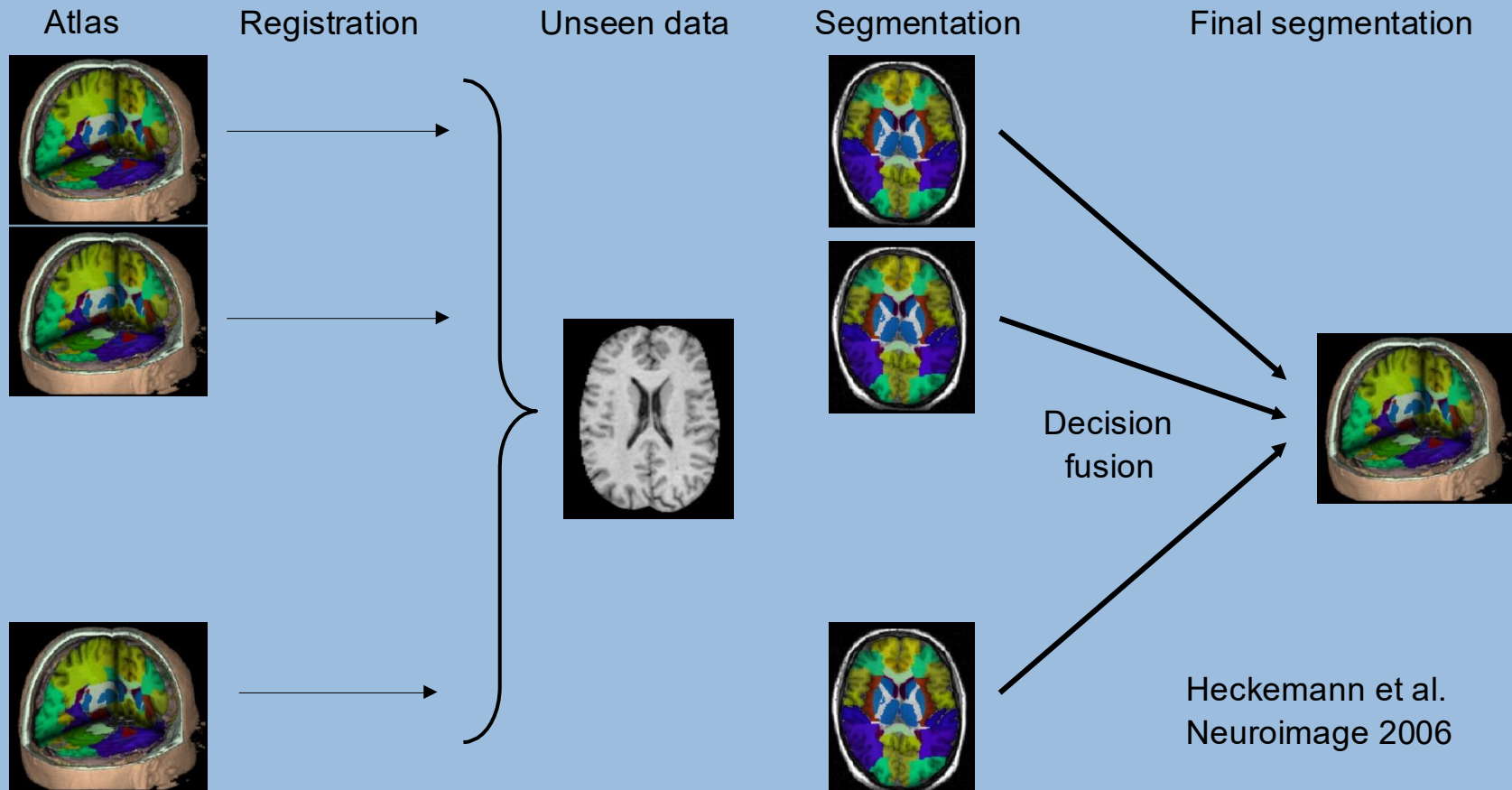
Multi-Atlas based segmentation



- T1-weighted MR images from 30 volunteers (age range 20-54 years, median age 30.5 years, 15 male, 15 female)
- 256 x 256 x 124 volumes (resolution $1.25 \times 0.94 \times 1.50$ mm)
- Each image is manually segmented in 83 anatomical structures

A. Hammers *et al.* Three-dimensional maximum probability atlas of the human brain, with particular reference to the temporal lobe. *Human Brain Mapping*, 19(4):224-247, 2003.

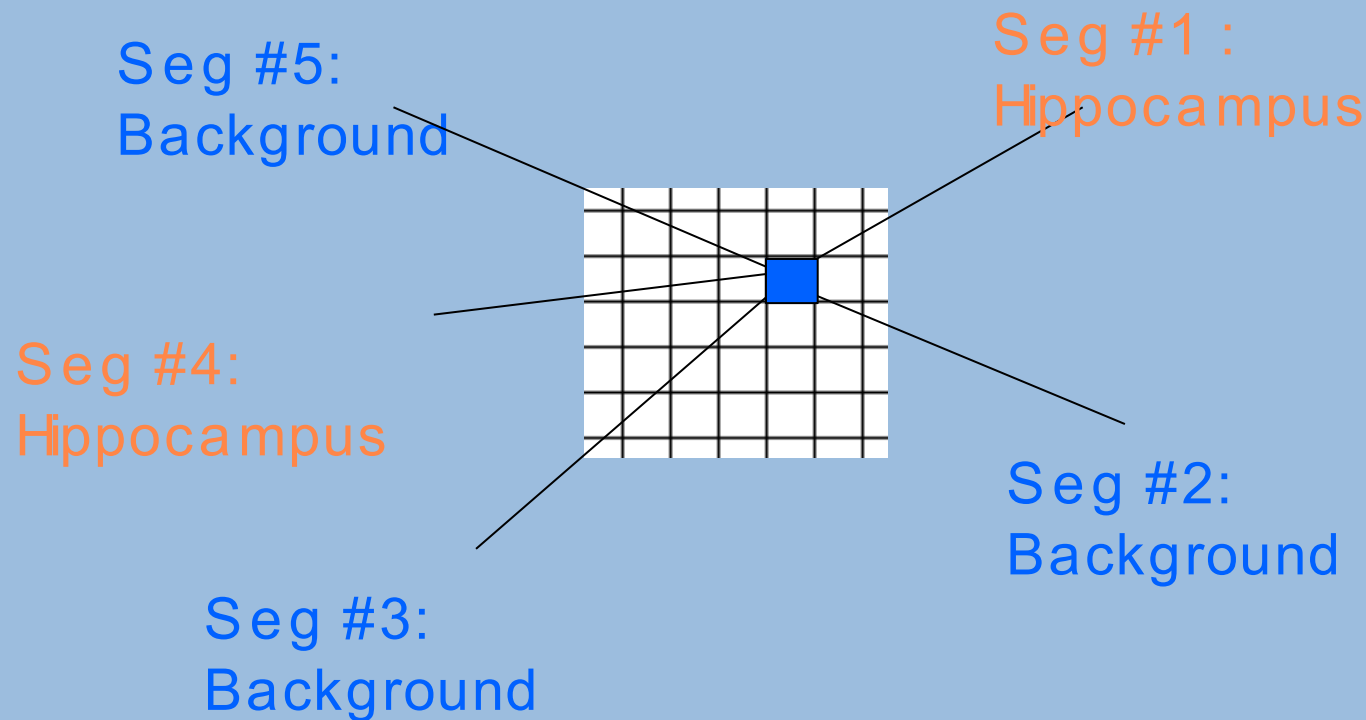
Multi-atlas segmentation using classifier fusion



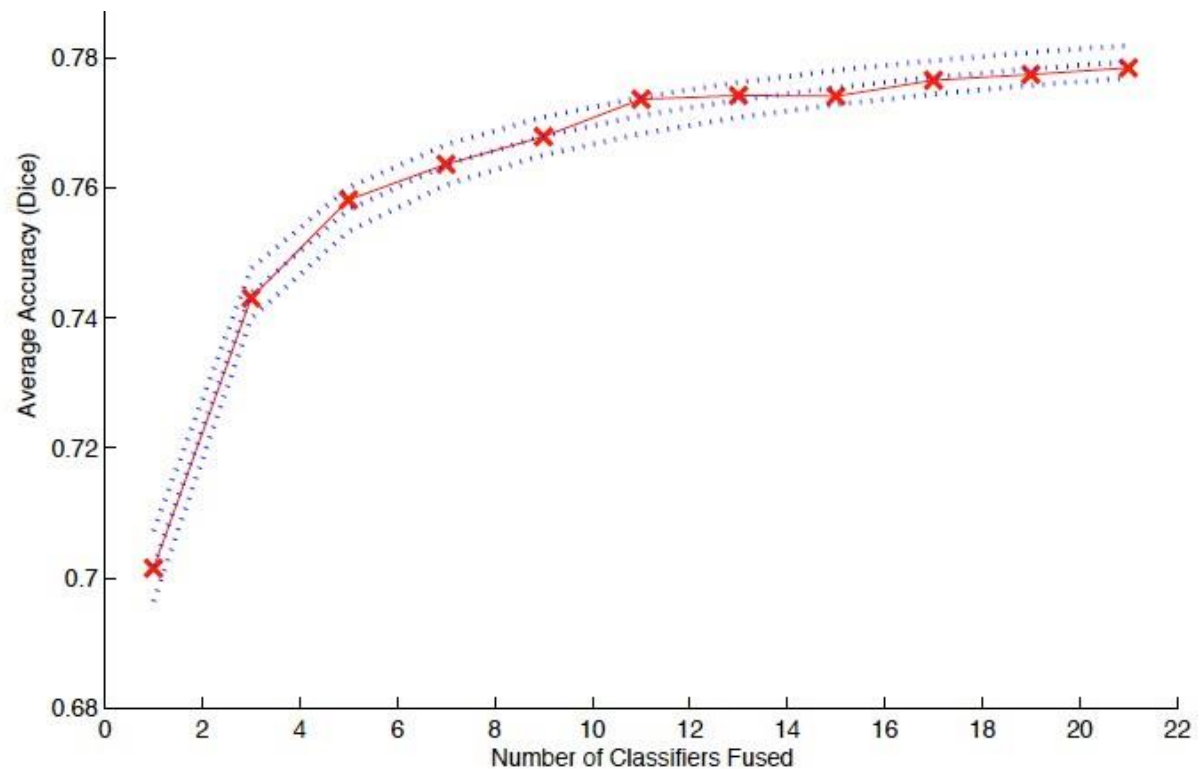
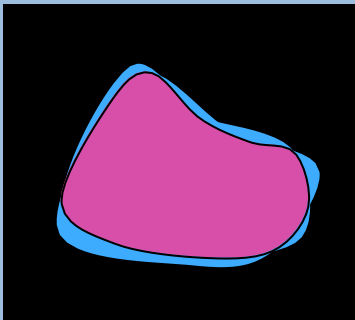
How do you fuse?

- Global fusion strategies:
 - Majority voting
 - Weighted voting
- Local fusion strategies:
 - Locally weighted fusion
 - Simultaneous Truth And Performance Level Estimation (STAPLE)
- Shape Based Averaging

Combining multiple segmentations: Simple majority voting



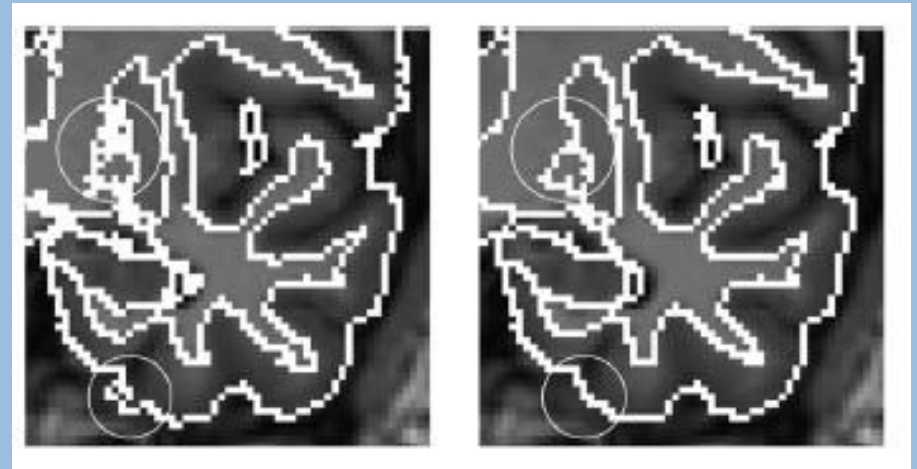
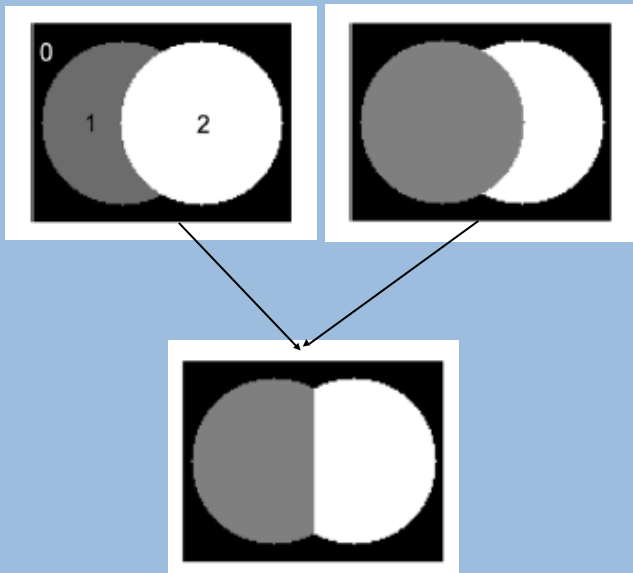
Multi-atlas segmentation using classifier fusion



Heckemann et al.
Neuroimage 2006

Combining multiple segmentations: Shape- based averaging

- Average the shapes of the segmentations
 - Based on averaging the distance transforms of the segmentations



VOTE

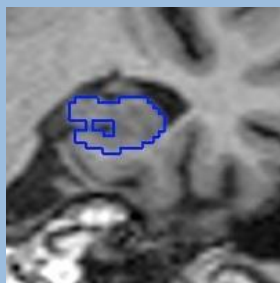
SBA

Combining multiple segmentations: STAPLE

Probabilistic approach



What is the **most probable** underlying true segmentations when you see the above segmentations?

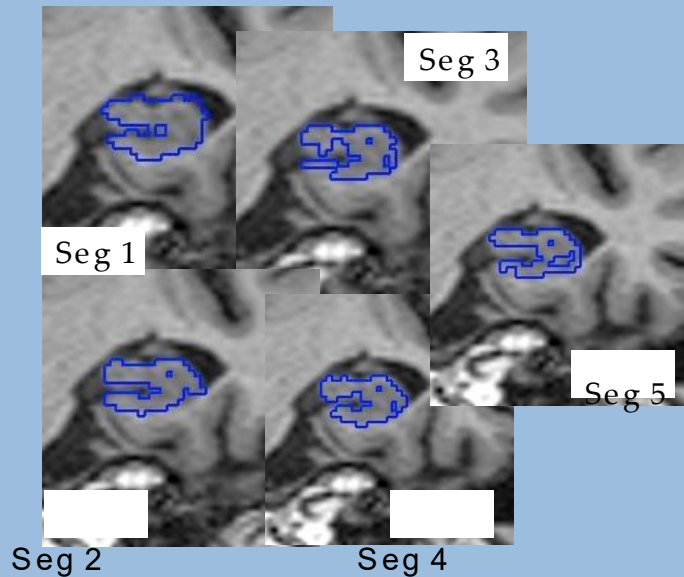


- At the same time, it tells you the performance (sensitivity and specificity) of each segmentation.
- Sensitivity p : true positive fraction; Specificity q : true negative fraction

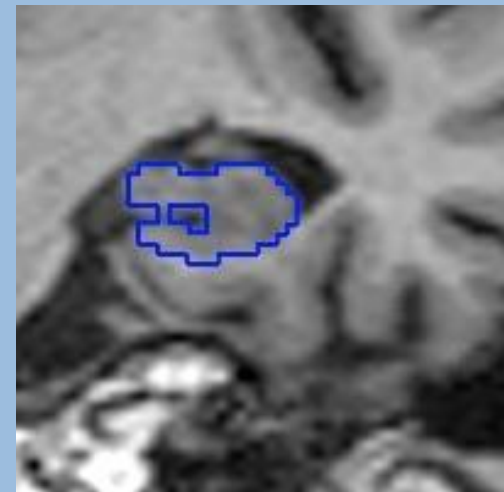
Combining multiple segmentations: STAPLE

- Can we find the performance of each segmentation (p, q) and the ground truth (T) that maximises the likelihood of observing the segmentations (D)?

D (Data/expert segmentations we want to merge) known



T (the ground Truth) unknown

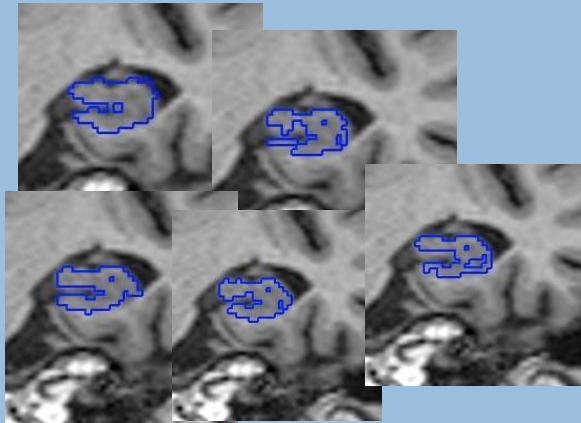


p, q (performance of each segmentation),
unknown

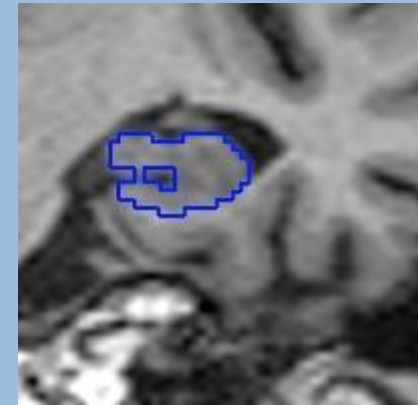
Combining multiple segmentations: STAPLE

Solved iteratively by Expectation-Maximization (EM) (Warfield et al., 2004)

D (Data/expert segmentations you want to merge)
known



T (the ground Truth)
unknown



p, q (performance of each segmentation),
unknown

$p^{(0)}, q^{(0)}$: initial estimate

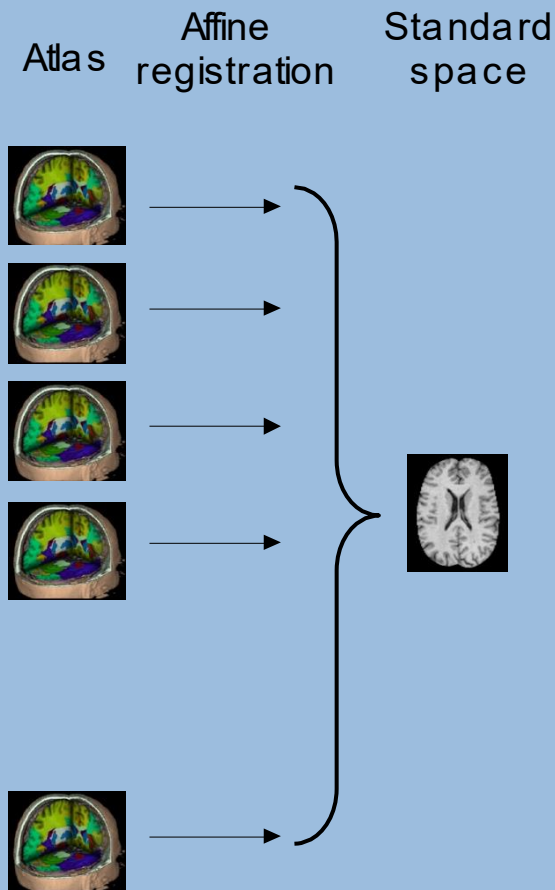
$p^{(1)}, q^{(1)}$: next estimate

$$f(T | D, p^{(0)}, q^{(0)})$$

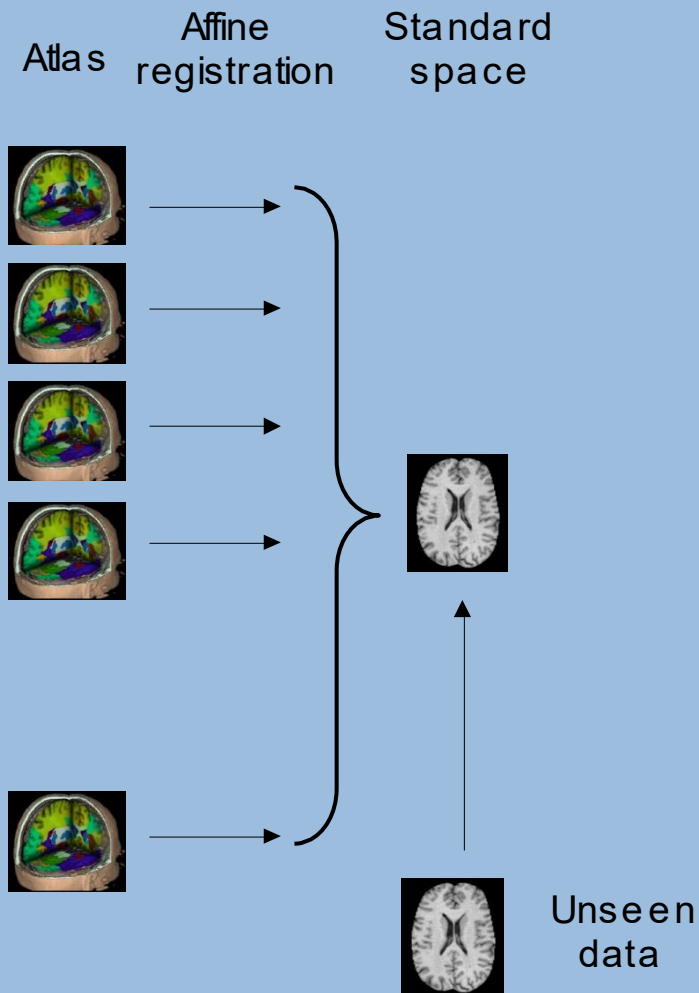
E-step

M-step : use the current ground truth to estimate the next $p^{(1)}, q^{(1)}$.

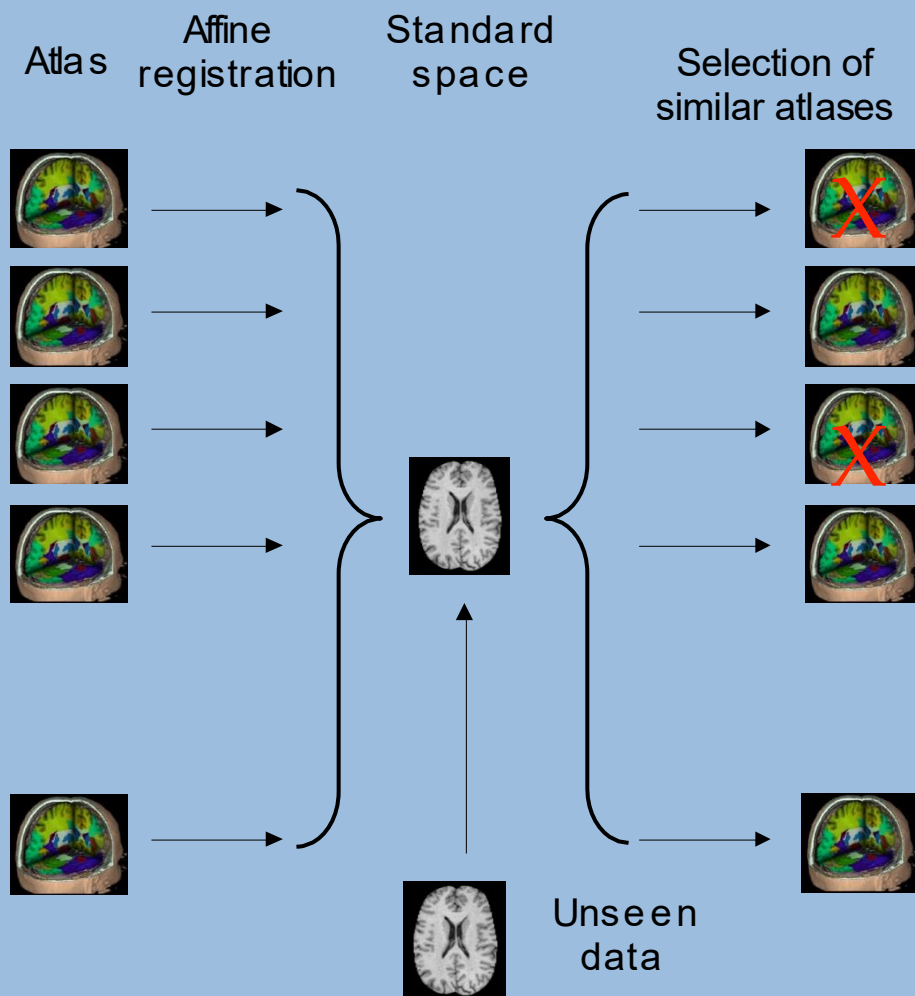
Multi-atlas segmentation using classifier fusion and selection



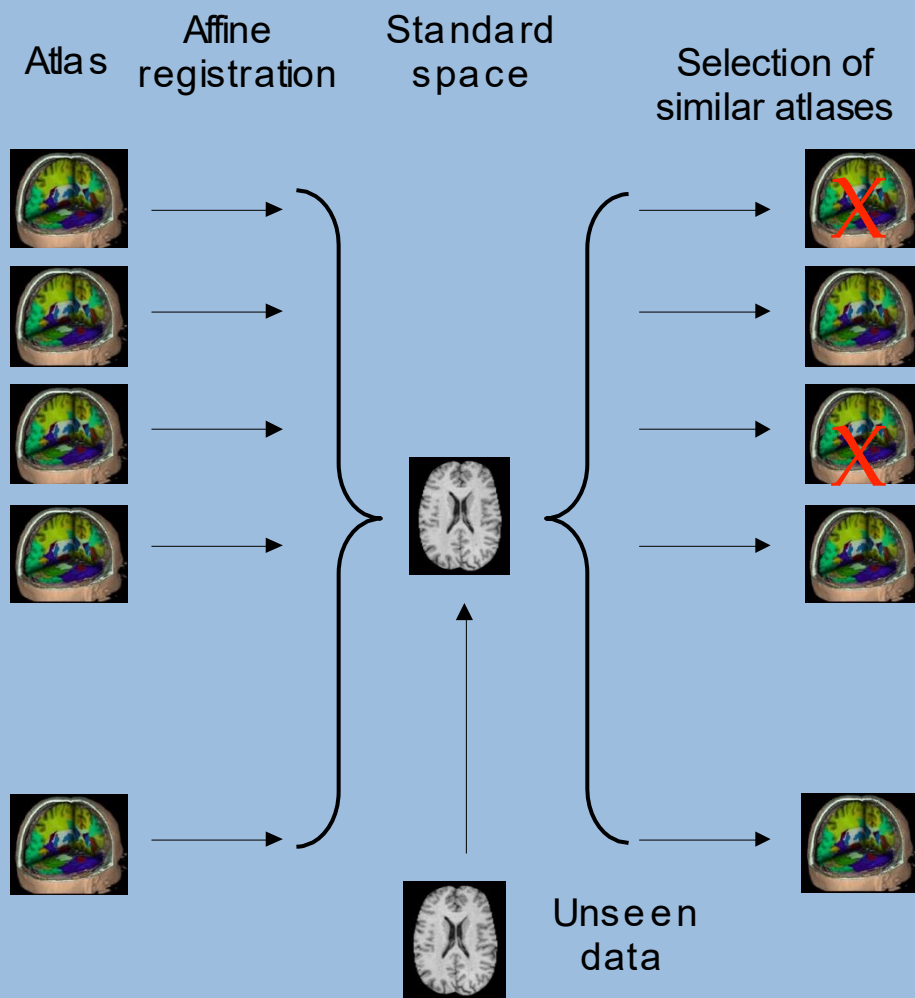
Multi-atlas segmentation using classifier fusion and selection



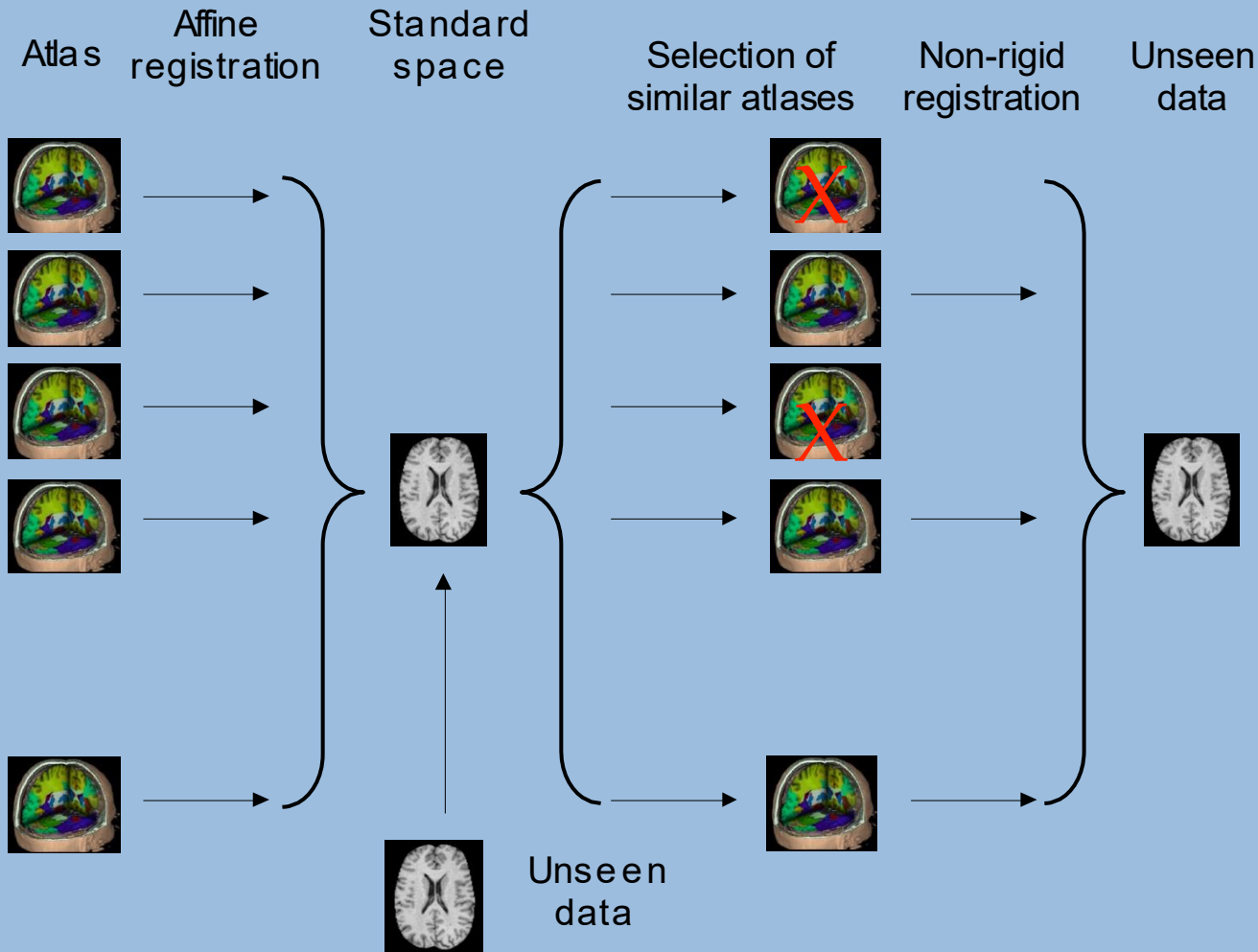
Multi-atlas segmentation using classifier fusion and selection



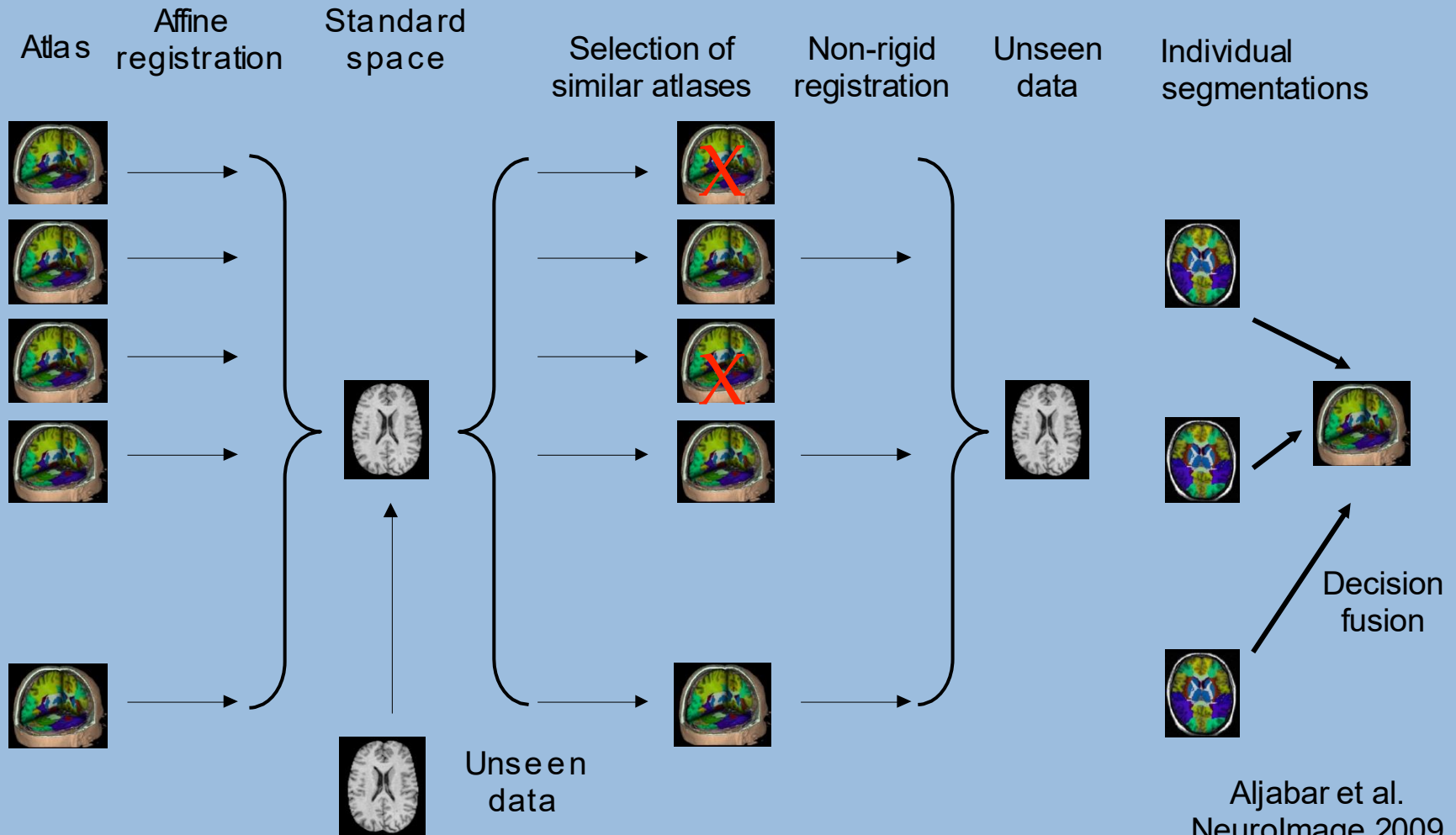
Multi-atlas segmentation using classifier fusion and selection



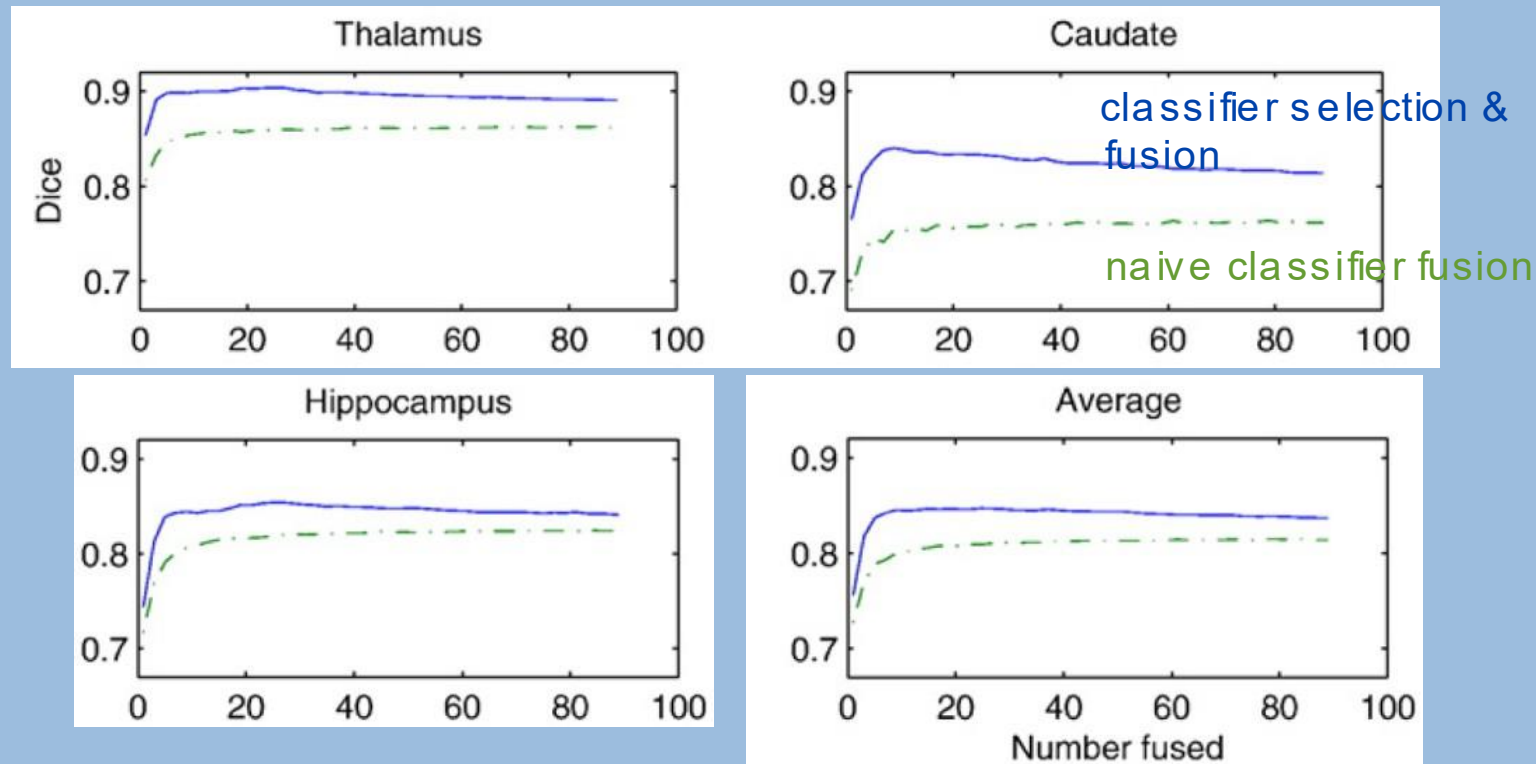
Multi-atlas segmentation using classifier fusion and selection



Multi-atlas segmentation using classifier fusion and selection



Multi-atlas segmentation using classifier fusion and selection



Patch-based segmentation

- Patch-based methods have been popular in the computer vision community, e.g.
 - texture synthesis (Efros and Freeman, 2001)
 - in-painting (Criminisi et al., 2004)
 - restoration (Buades et al., 2005)
 - single-frame super resolution (Protter et al., 2009).
- Basic idea:
 - only coarse registration required (e.g. rigid or affine registration)
 - use the same concept as label fusion

Patch-based denoising

- Buades et al. 2005
 - uses a neighborhood averaging strategy called non-local means (NLM)
 - assumes that every patch in a natural image has many similar patches in the same image.

$$\forall \mathbf{x} \in \Omega, I_{nlm}(\mathbf{x}) = \frac{\sum_{\mathbf{y} \in \Omega} w(\mathbf{x}, \mathbf{y}) I(\mathbf{y})}{\sum_{\mathbf{y} \in \Omega} w(\mathbf{x}, \mathbf{y})}$$

Denoised image

Self-similarity

Original image

Patch-based denoising

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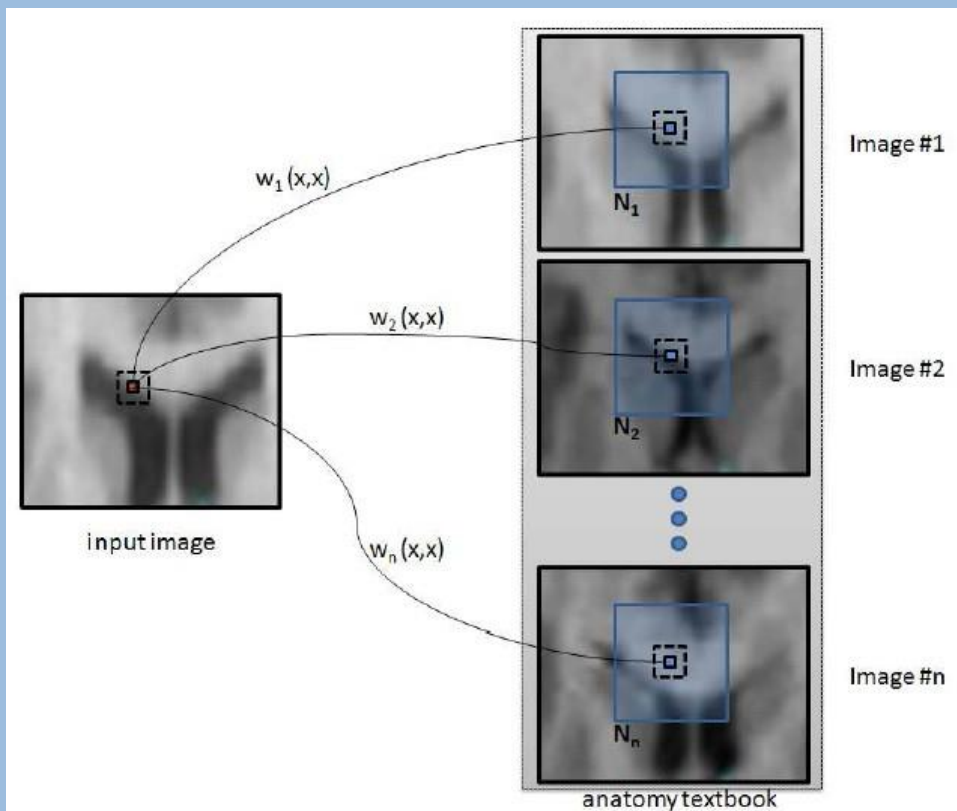
$$w(\mathbf{x}, \mathbf{y}) = f \left(\frac{\sum_{\mathbf{x}' \in \mathcal{P}_I(\mathbf{x}), \mathbf{y}' \in \mathcal{P}_I(\mathbf{y})} (I(\mathbf{x}') - I(\mathbf{y}'))^2}{2N\beta\hat{\sigma}^2} \right)$$

Weighting function
usually: $\exp(-x)$

Bandwidth for smoothing

Patch-based denoising

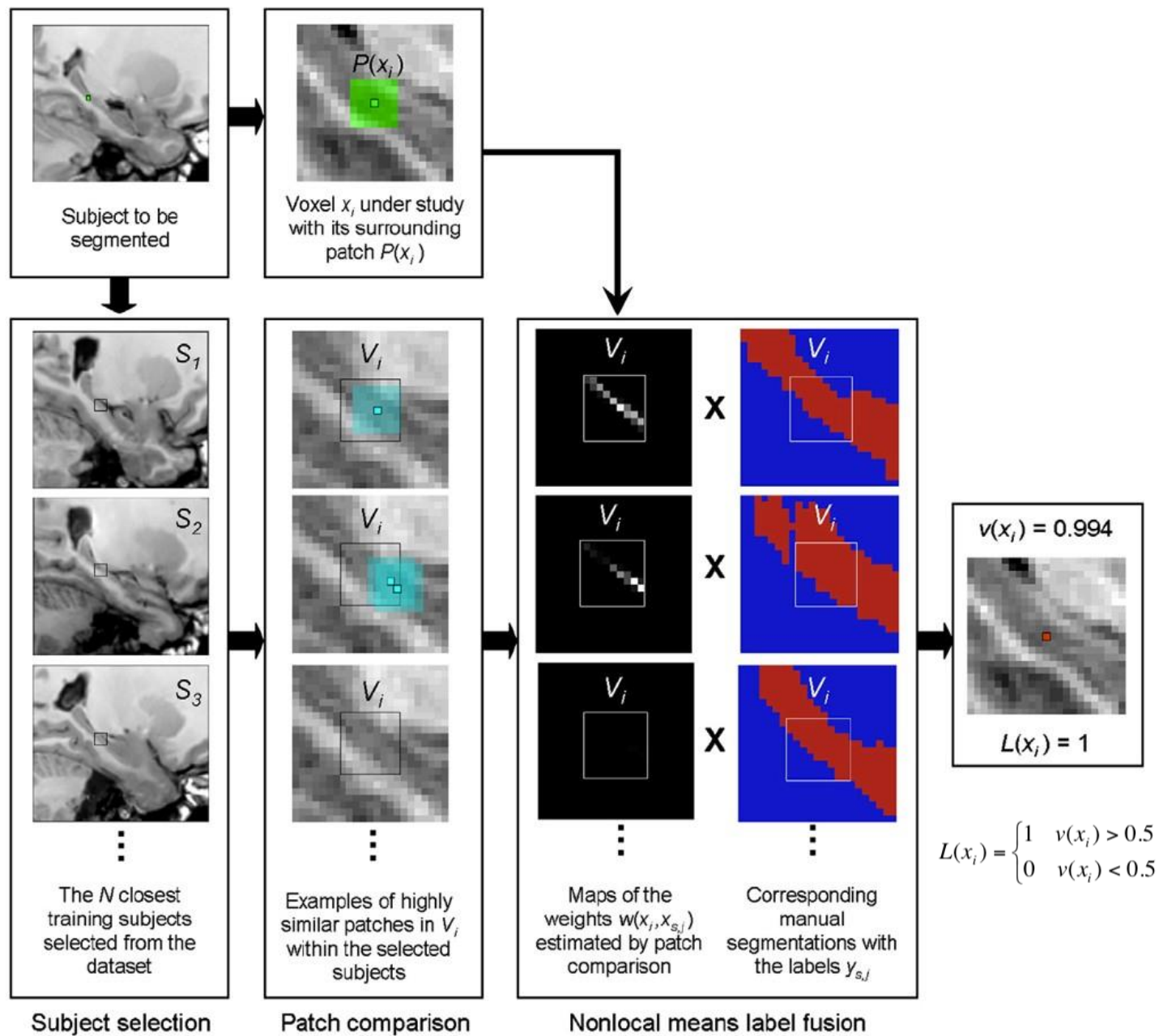
- Apply the same idea as in denoising to label fusion:
 - Simultaneously proposed by Rousseau et al. and Coupe et al. in 2011



$$w_i(x, y) = f \left(\frac{\sum_{x' \in \mathcal{P}_I(x), y' \in \mathcal{P}_{\mathcal{I}_i}(y)} (I(x') - \mathcal{I}_i(y'))^2}{2N\beta\hat{\sigma}^2} \right)$$

$$\forall x \in \Omega, L(x) = \frac{\sum_{i=1}^n \sum_{y \in \mathcal{N}(x)} w_i(x, y) \mathcal{L}_i(y)}{\sum_{i=1}^n \sum_{y \in \mathcal{N}(x)} w_i(x, y)}$$

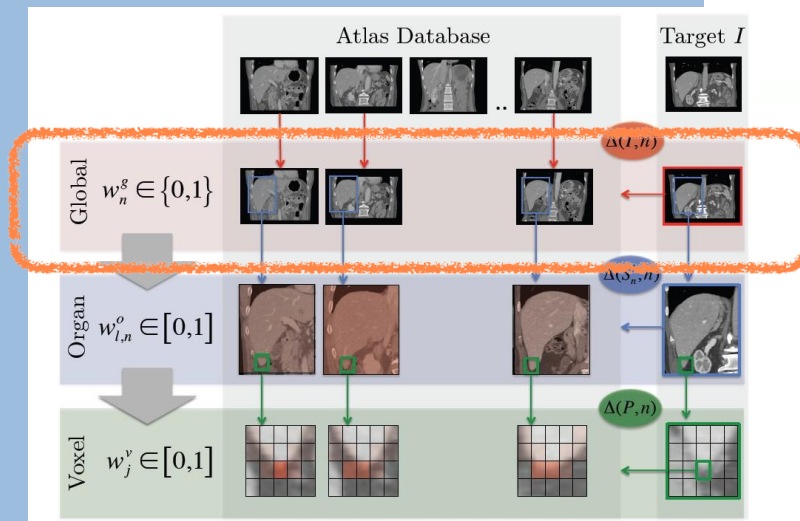
Rousseau et al. 2011



Patch-based multi-organ segmentation of abdominal CT

- Segmentation of CT scans from 150 subjects
- Hierarchical, patch-based label fusion process

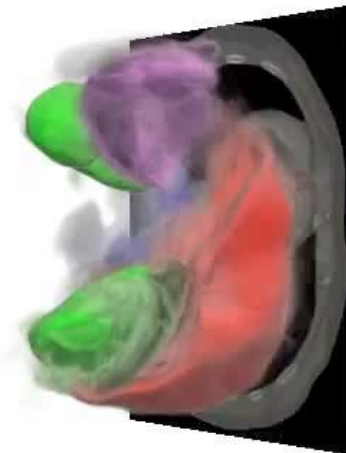
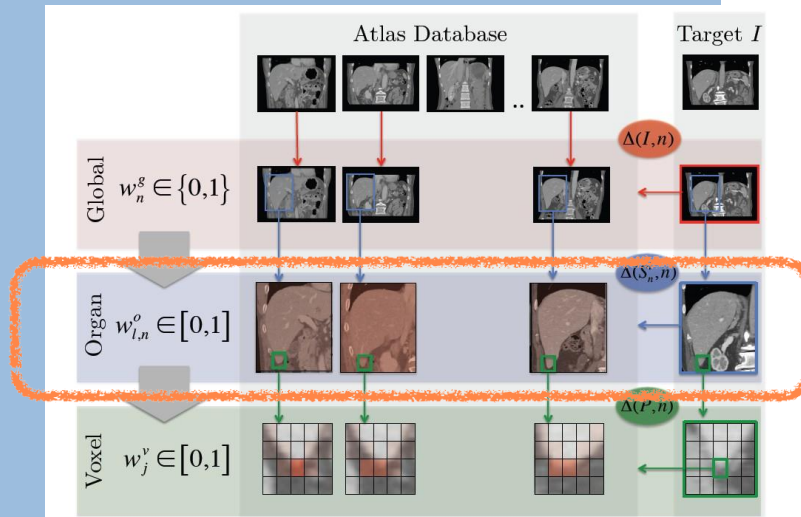
Structure	Dice	Jaccard
Kidneys	92.5 (7.2) [51.5 98.2]	86.8 (10.5) [34.6 96.4]
Liver	94.0 (2.8) [81.4 97.3]	88.9 (4.8) [68.7 94.9]
Pancreas	69.6 (16.7) [6.9 90.0]	55.5 (17.1) [3.6 83.3]
Spleen	92.0 (9.2) [26.4 98.2]	86.2 (12.7) [15.2 96.4]



Patch-based multi-organ segmentation of abdominal CT

- Segmentation of CT scans from 150 subjects
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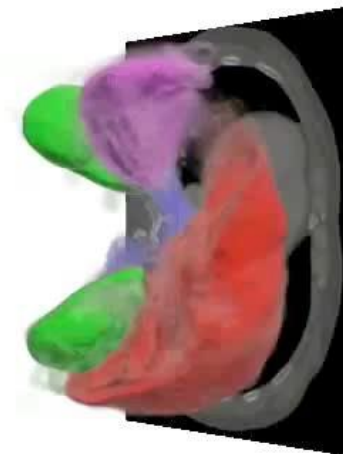
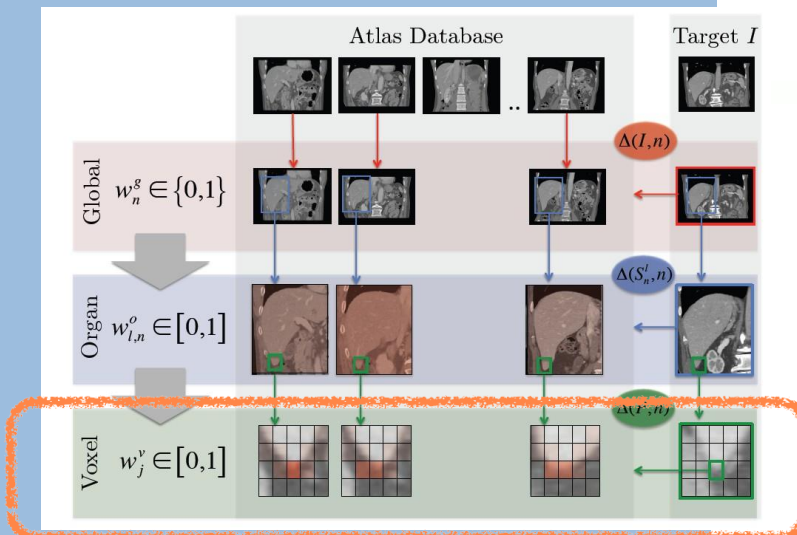
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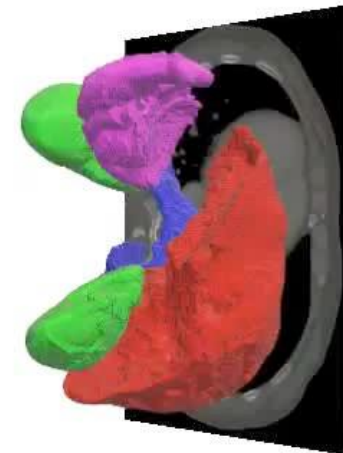
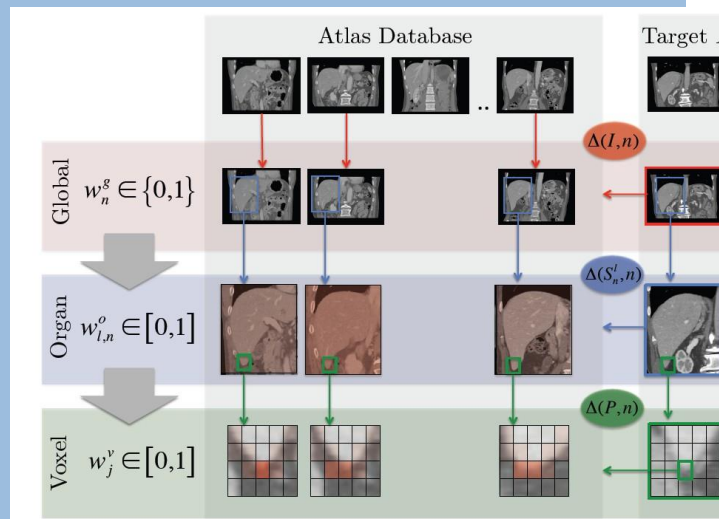
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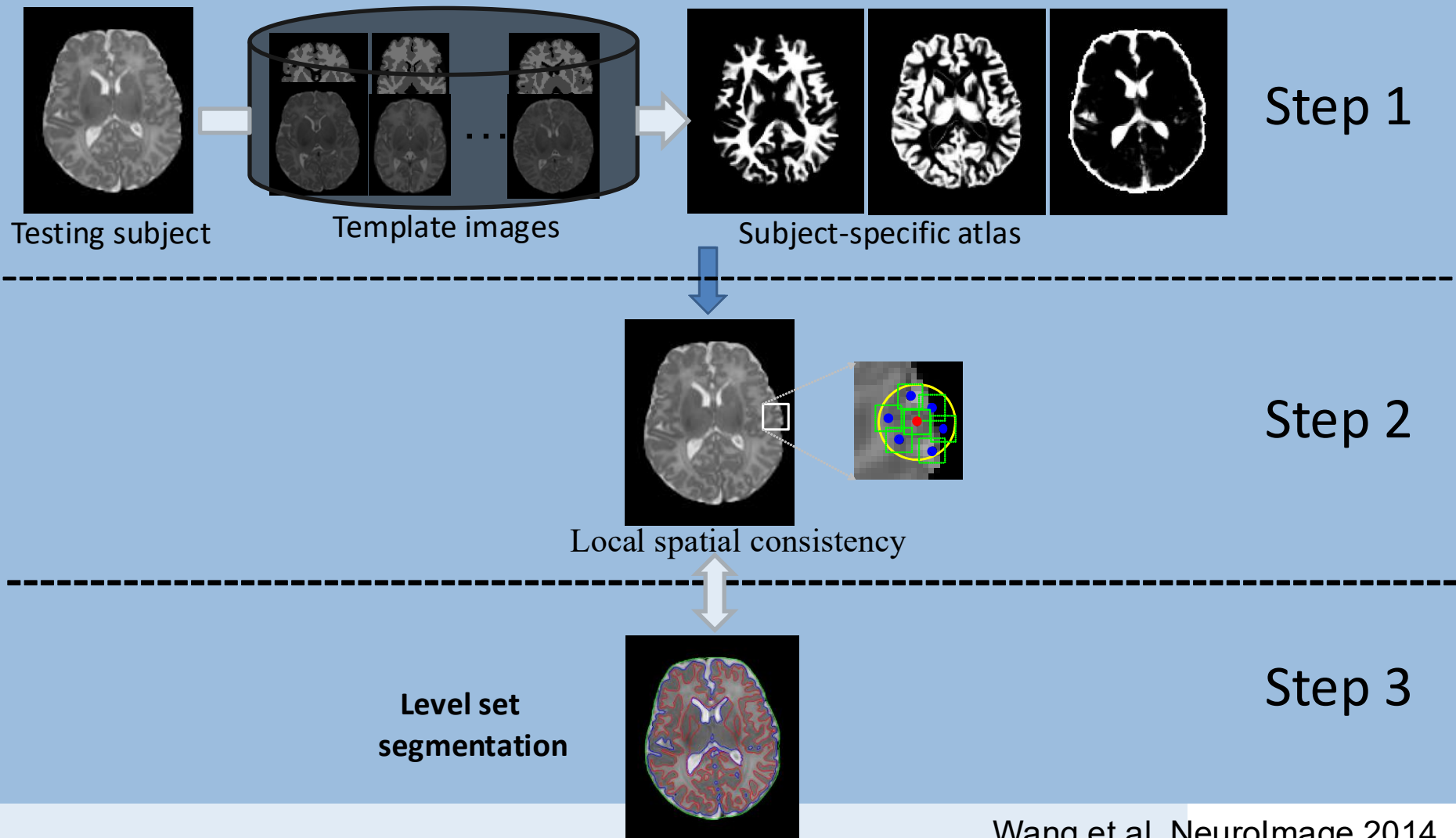
Patch-based multi-organ segmentation of abdominal CT

- Segmentation of CT scans from 150 subjects
- Hierarchical, patch-based label fusion process **plus graph-cut refinement**

Structure	Dice	Jaccard
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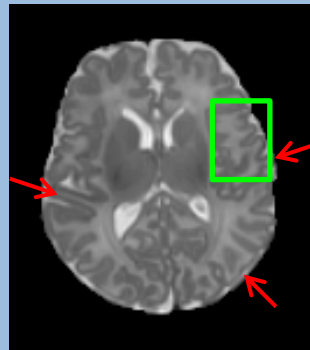


Patch-based multi-organ segmentation of brain structures

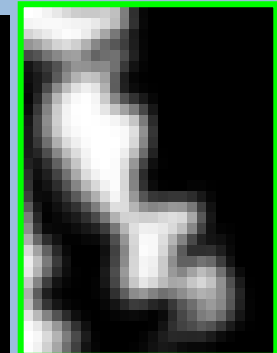
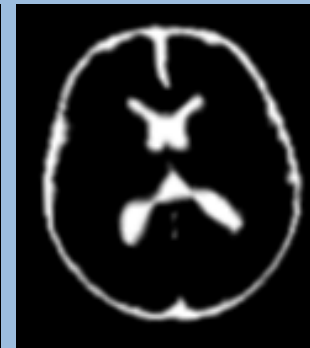
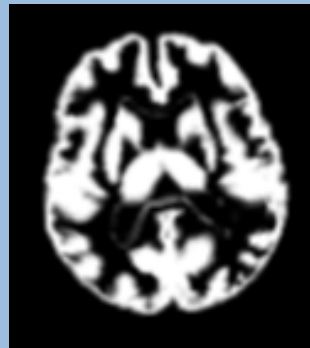
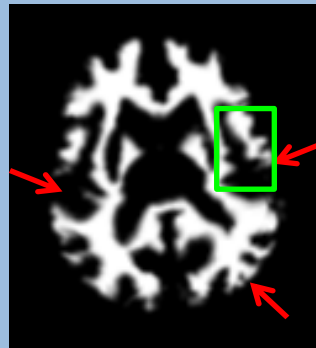


Patch-based multi-organ segmentation of brain structures

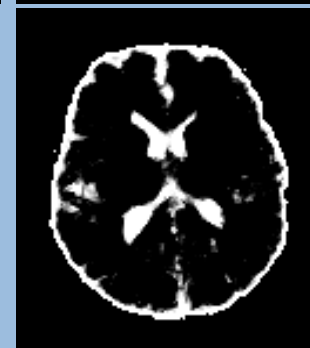
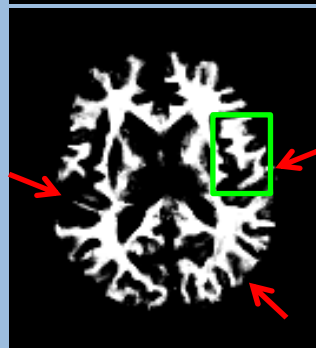
Original T2
image



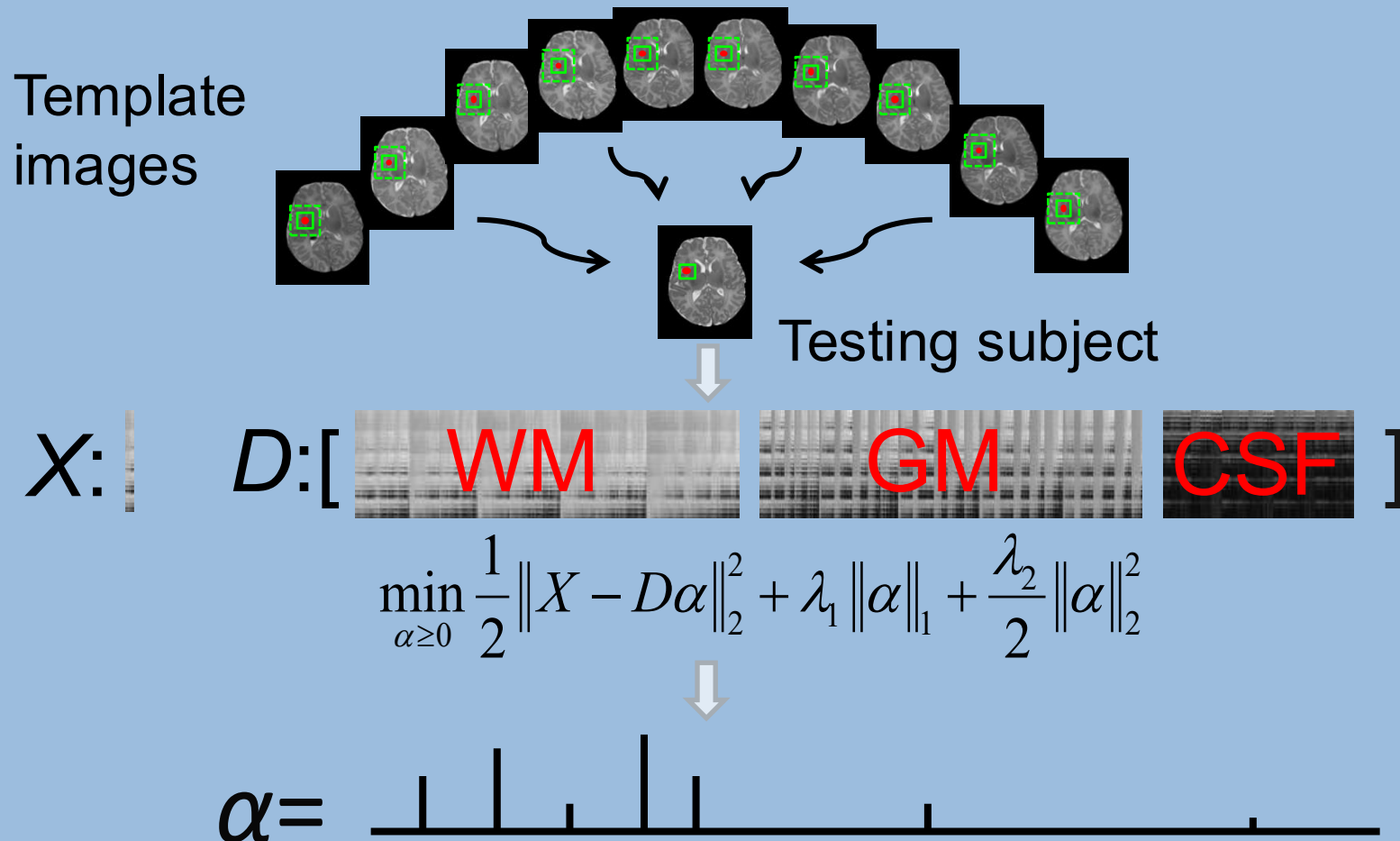
Population-
based atlas



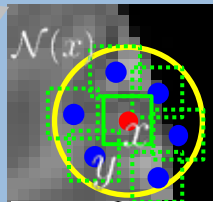
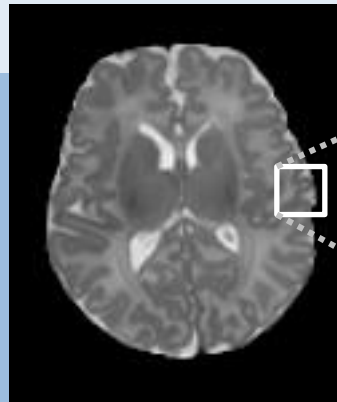
Subject-specific
atlas



Patch-based multi-organ segmentation of brain structures



Patch-based multi-organ segmentation of brain structures



$$P_j^{Patch_L}(x) \propto \frac{1}{Z} \sum_{y \in \mathcal{N}(x)} \underbrace{\alpha'_y}_{\text{patch similarity}} \underbrace{p_{j,y}(I(x))}_{LGD}$$

$$\min_{\alpha' \geq 0} \frac{1}{2} \|X - D'\alpha'\|_2^2 + \lambda_1 \|\alpha'\|_1 + \frac{\lambda_2}{2} \|\alpha'\|_2^2$$

$$p_{j,y}(I(x)) = \frac{1}{\sqrt{2\pi}\sigma_j(y)} \exp\left(-\frac{(u_j(y) - I(x))^2}{2\sigma_j^2(y)}\right)$$

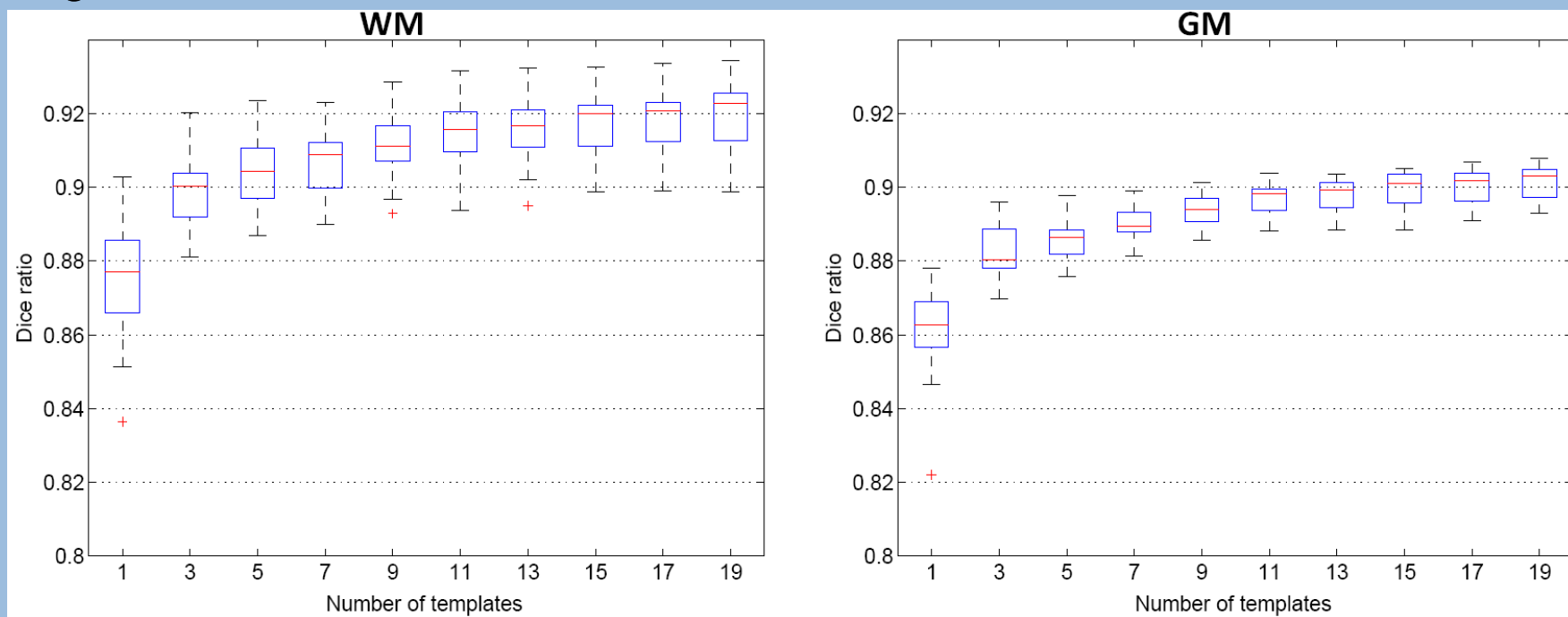
$$P_j^{Patch_L_Prior}(x) = \underbrace{P_j(x)}_{\text{prior}} P_j^{Patch_L}(x)$$

Step 1: subject-specific atlas

$$\varepsilon = - \sum_{j=1} \int_{x \in \Omega_j} \log P_j^{Patch_L_Prior}(x) dx$$

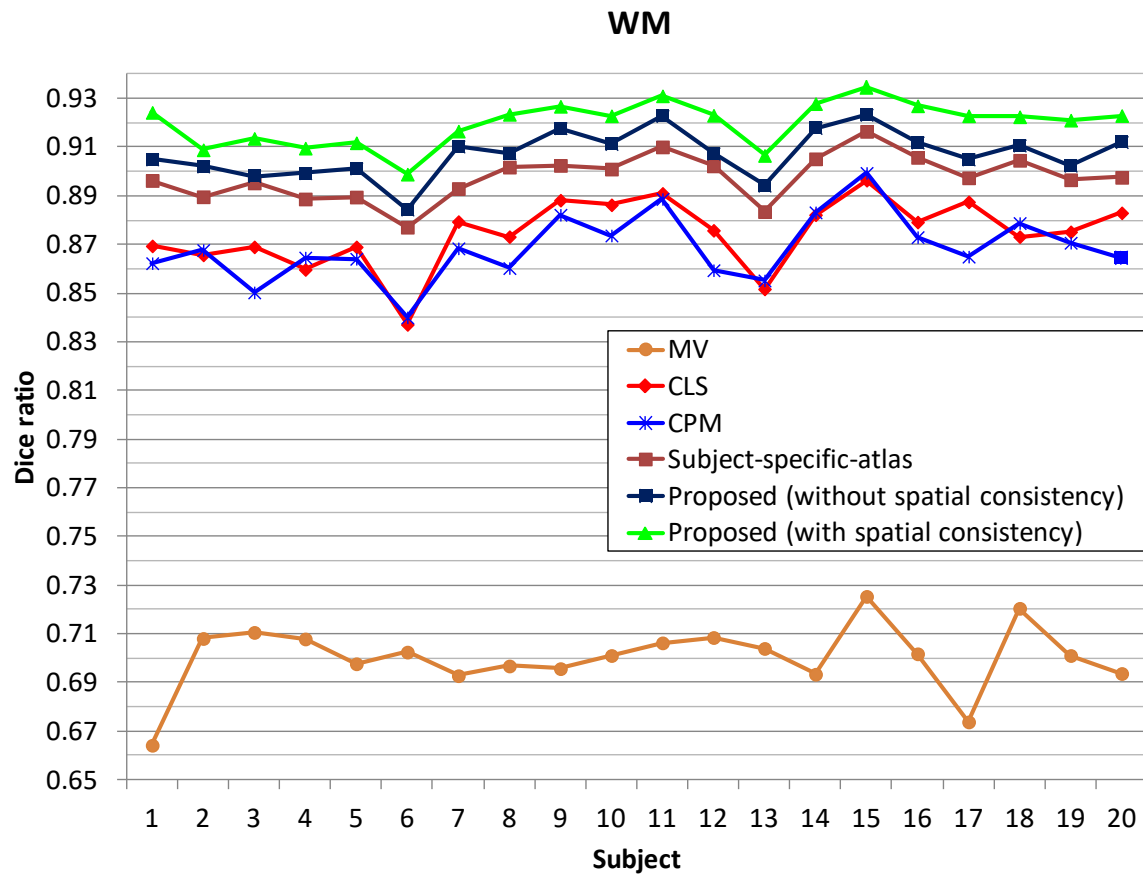
Template numbers?

How many template images are needed to generate a good segmentation?



Box-whisker plots of Dice ratio of segmentation using an increasing number of templates from the library. Experiment is performed by leave-one-out using the library of 20 templates.

Leave-one-out cross validation on 20 subjects

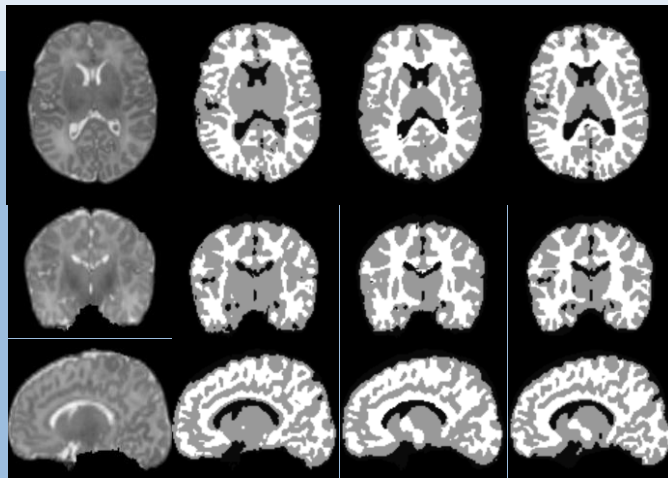


MV: Majority voting

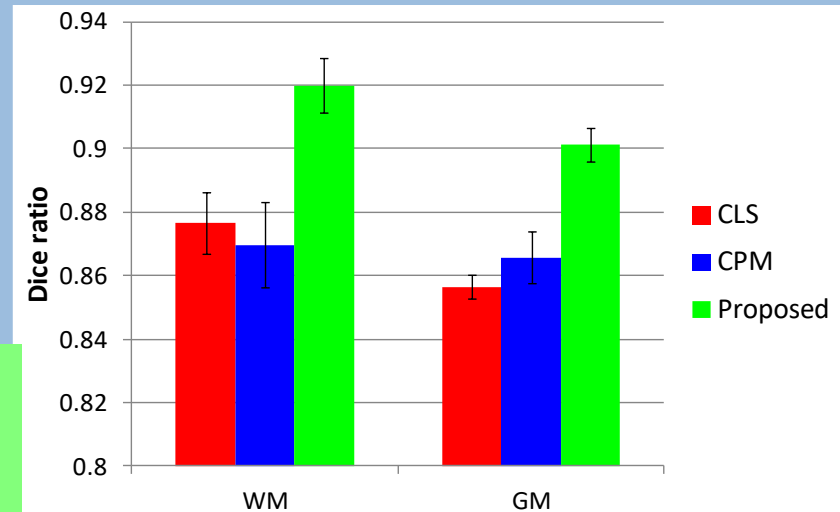
CLS (Coupled level sets): Wang, L., et al., 2011. NeuroImage

CPM (Conventional patch-based method): Coupe, P., et al., 2011. NeuroImage

8 testing subjects with manual segmentations

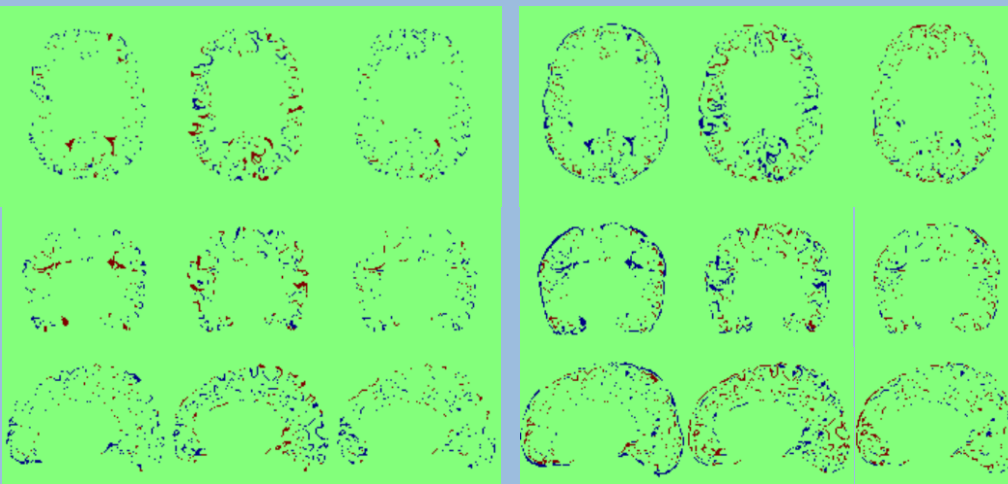


(a) Original (b) CLS (c) CPM (d) Proposed



CLS: Coupled level set

CPM: Conventional patch-based method



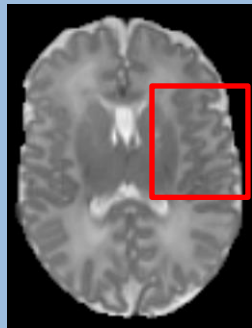
(e) CLS (f) CPM (g) Proposed

WM difference

(h) CLS (i) CPM (j) Proposed

GM difference

94 testing subjects for qualitative evaluation



Original



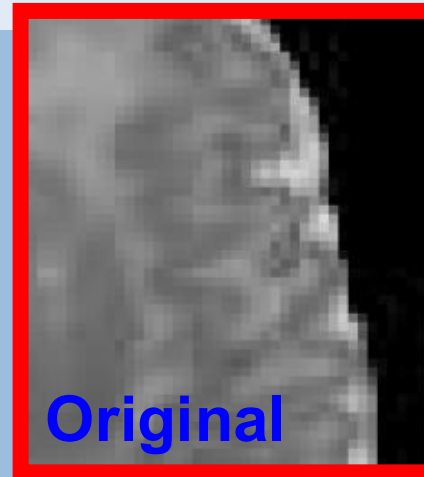
CLS



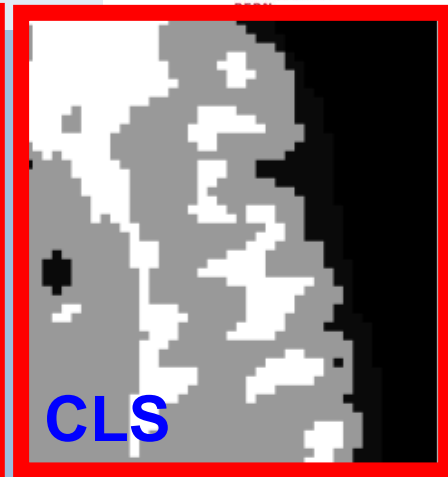
CPM



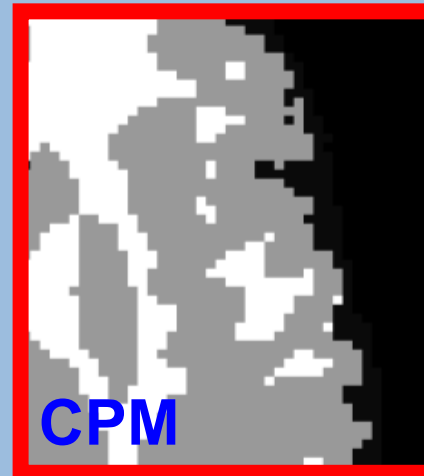
Proposed



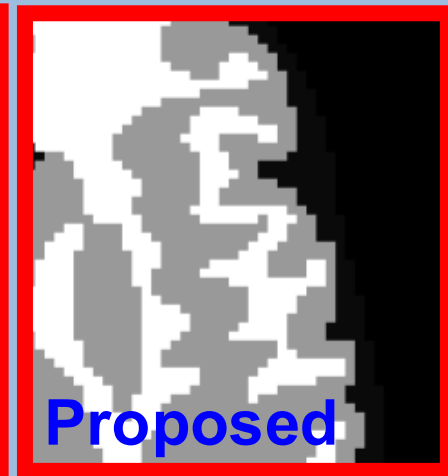
Original



CLS



CPM

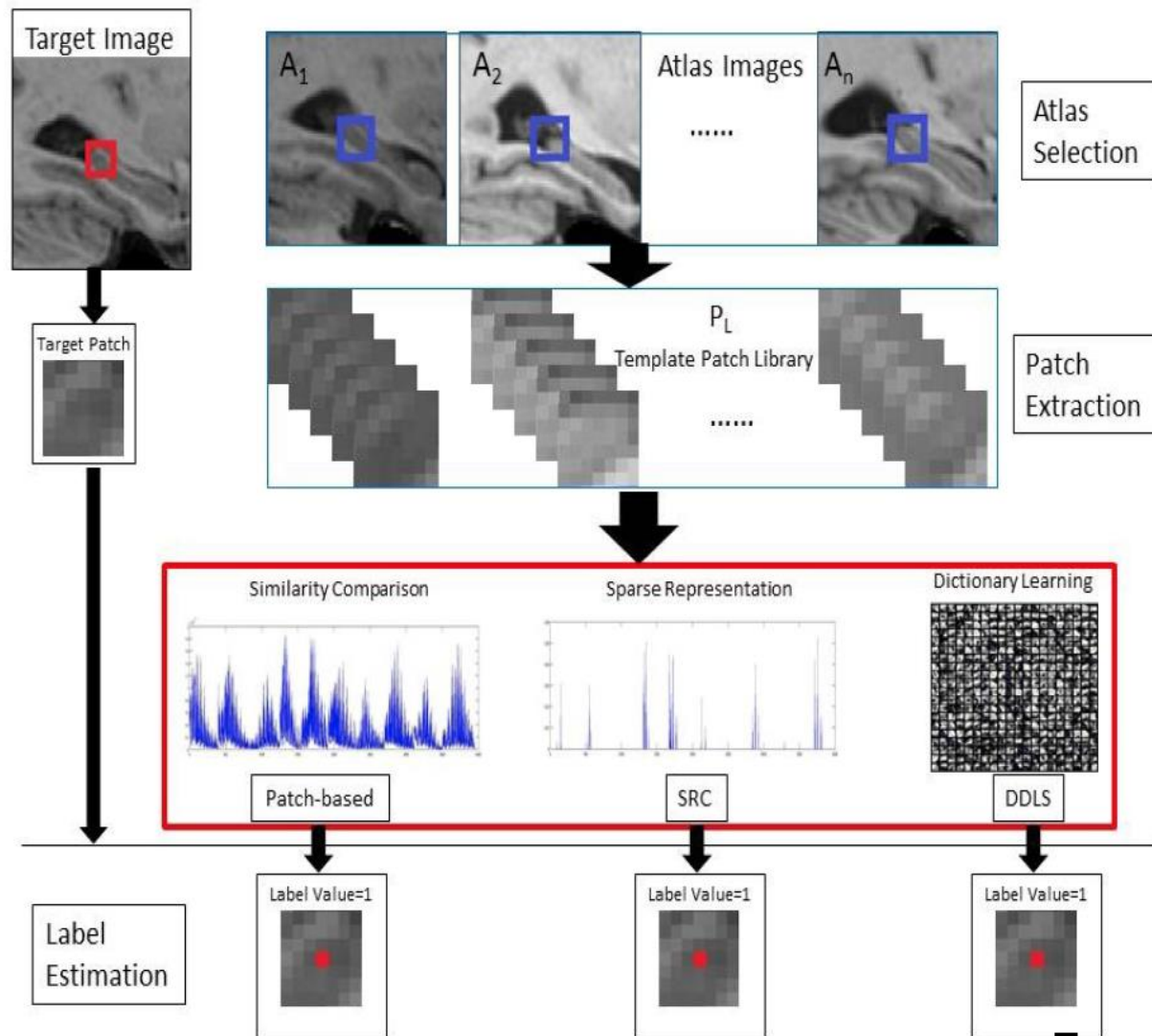


Proposed

CLS: Coupled level set

CPM: Conventional patch-based method

Patch-based Labeling vs. Sparse Representation Classification (SRC) vs. Discriminative Dictionary Learning for Segmentation (DDLs)



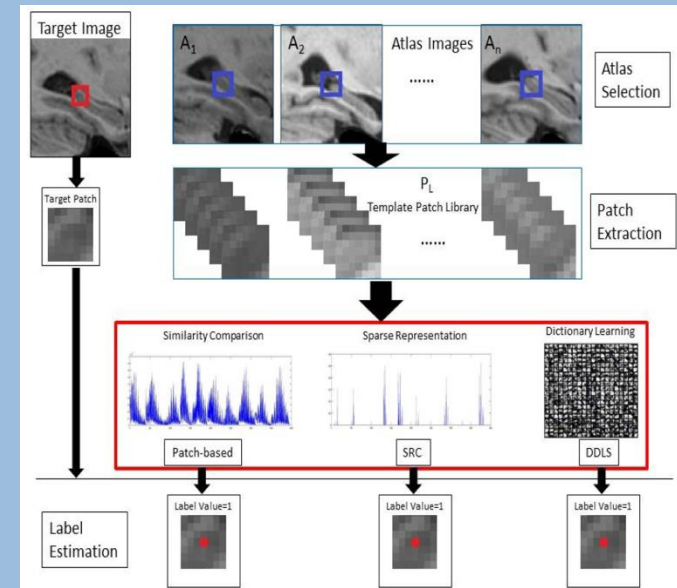
Patch-based Labeling vs. Sparse Representation Classification (SRC) vs. Discriminative Dictionary Learning for Segmentation (DDLs)

- SRC assumes a target patch represented by a few representative patches
- DDLS adds a learned linear classifier

Target patch

$$p_t = a_1 p_1 + a_2 p_2 + \dots + a_n p_n$$

a_i : coefficient; sparse
 p_i : patch library

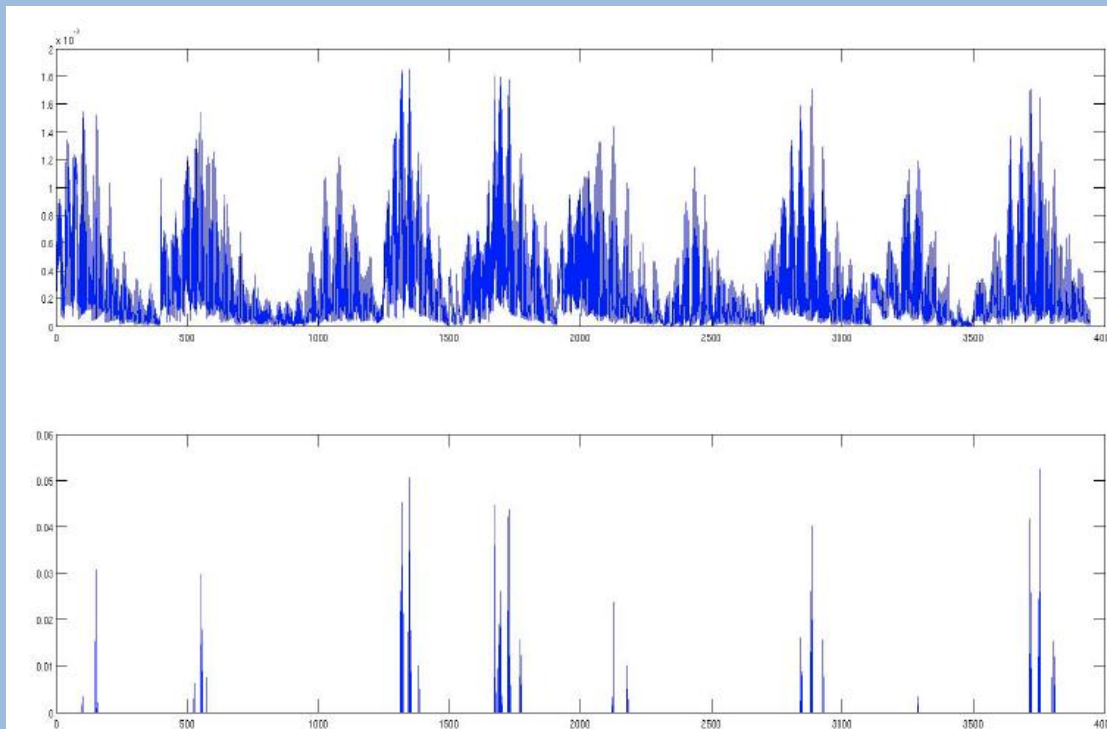


$$\hat{a} = \min_a \frac{1}{2} \|p_t - P_L a\|_2^2 + \lambda_1 \|a\|_1 + \frac{\lambda_2}{2} \|a\|_2^2$$

Elastic-Net problem

Patch-based Labeling vs. Sparse Representation Classification (SRC) vs. Discriminative Dictionary Learning for Segmentation (DDLs)

- SRC assumes a target patch represented by a few representative patches



Patch-based

SRC

Patch-based Labeling vs. Sparse Representation Classification (SRC) vs. Discriminative Dictionary Learning for Segmentation (DDLS)

- DDLS:
 - Decrease computational burden via smaller-size dictionary
 - Better exploit discriminative power of patch library

Reconstruction error

Classification accuracy

$$\langle D, W, \alpha \rangle = \arg \min_{D, W, \alpha} \boxed{\|P_L - D\alpha\|_2^2} + \beta_1 \boxed{\|H - W\alpha\|_2^2}$$

subject to $\|\alpha\|_0 \leq T$

Patch-based Labeling vs. Sparse Representation Classification (SRC) vs. Discriminative Dictionary Learning for Segmentation (DDLs)

- DDLs:
 - Decrease computational burden via smaller-size dictionary
 - Better exploit discriminative power of patch library

$$\langle D, W, \alpha \rangle = \arg \min_{D, W, \alpha} \|P_L - D\alpha\|_2^2 + \beta_1 \|H - W\alpha\|_2^2$$

subject to $\|\alpha\|_0 \leq T$

$$\langle D, W, \alpha \rangle = \arg \min_{D, W, \alpha} \left\| \begin{pmatrix} P_L \\ \sqrt{\beta_1} H \end{pmatrix} - \begin{pmatrix} D \\ \sqrt{\beta_1} W \end{pmatrix} \alpha \right\|_2^2$$

subject to $\|\alpha\|_0 \leq T$

Grouping patch and
label information

$$\langle \tilde{D}, \alpha \rangle = \arg \min_{\tilde{D}, \alpha} \left\| \tilde{P}_L - \tilde{D}\alpha \right\|_2^2 + \beta_2 \|\alpha\|_1$$

Impose sparsity

$$\tilde{D} = (D^t, \sqrt{\beta_1} W^t)^t, \tilde{P}_L = (P_L^t, \sqrt{\beta_1} H^t)^t$$

Patch-based Labeling vs. Sparse Representation Classification (SRC) vs. Discriminative Dictionary Learning for Segmentation (DDLs)

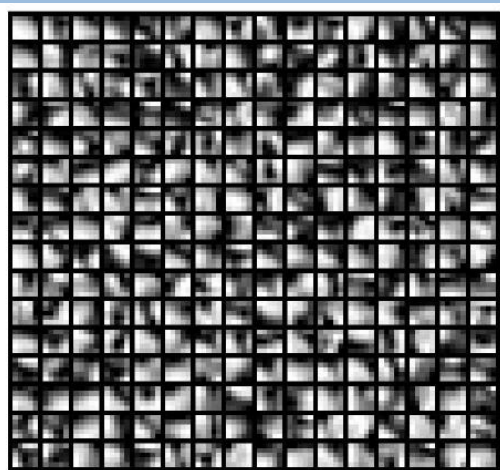
- DDLs:
 - Decrease computational burden via smaller-size dictionary
 - Better exploit discriminative power of patch library

$$\tilde{D}_t = \left\{ \begin{pmatrix} \tilde{d}_1 \\ \sqrt{\beta_1} \tilde{w}_1 \end{pmatrix}, \begin{pmatrix} \tilde{d}_2 \\ \sqrt{\beta_1} \tilde{w}_2 \end{pmatrix}, \dots, \begin{pmatrix} \tilde{d}_K \\ \sqrt{\beta_1} \tilde{w}_K \end{pmatrix} \right\}$$

Normalized voxel-wise
dictionaries and classifiers

$$\hat{D}_t = \left\{ \frac{\tilde{d}_1}{\|\tilde{d}_1\|_2}, \frac{\tilde{d}_2}{\|\tilde{d}_2\|_2}, \dots, \frac{\tilde{d}_K}{\|\tilde{d}_K\|_2} \right\} \quad \hat{W}_t = \left\{ \frac{\tilde{w}_1}{\|\tilde{d}_1\|_2}, \frac{\tilde{w}_2}{\|\tilde{d}_2\|_2}, \dots, \frac{\tilde{w}_K}{\|\tilde{d}_K\|_2} \right\}$$

Final {D,W}



Example of dictionary:
The dictionary has 16x16 atoms of size 53

Patch-based Labeling vs. Sparse Representation Classification (SRC) vs. Discriminative Dictionary Learning for Segmentation (DDLs)

- Three-steps for DDLs:

1.- Extract patch p

2. Solve for:

$$\hat{\alpha}_t = \arg \min_{\alpha_t} \left\| p_t - \hat{D}_t \alpha_t \right\|_2^2 + \beta_2 \|\alpha_t\|_1$$

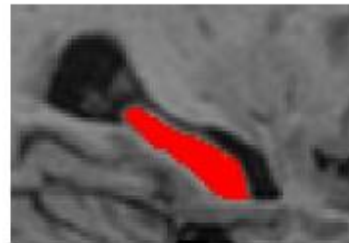
3.- Label voxel using classifier

$$\begin{cases} h_t = \hat{W}_t \hat{\alpha}_t \\ v = \arg \max_j h_t(j) \end{cases}$$

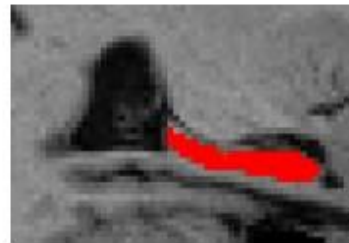
h_t : class label vector

Discriminative Dictionary Learning for Segmentation (DDLs)

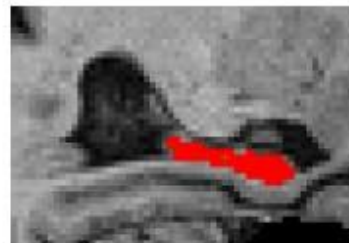
Manual segmentations



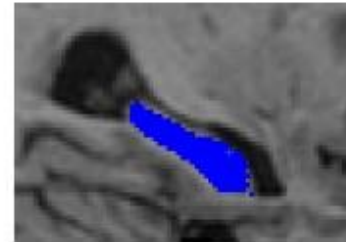
Best subject



Median subject



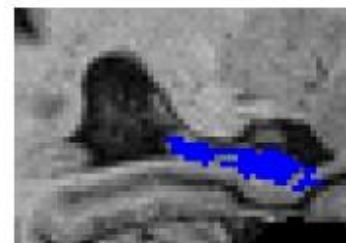
Worst subject



$k = 0.8767$



$k = 0.8233$

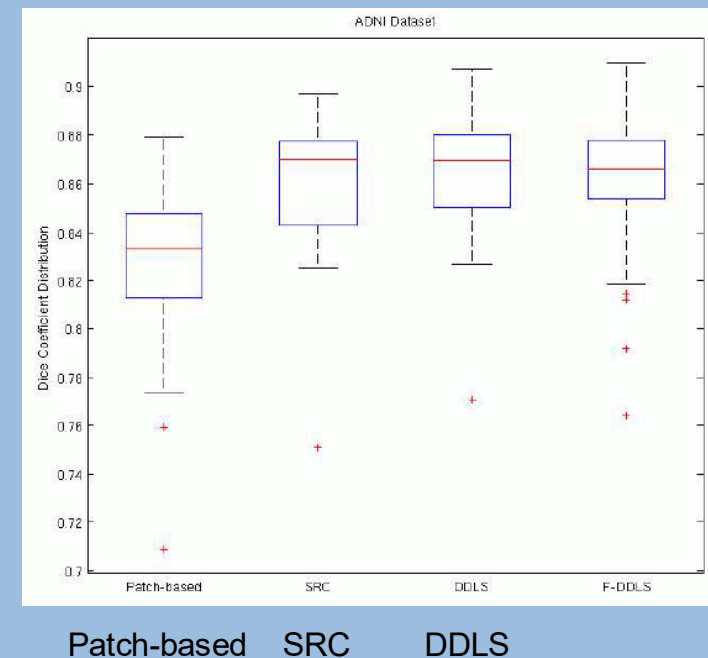
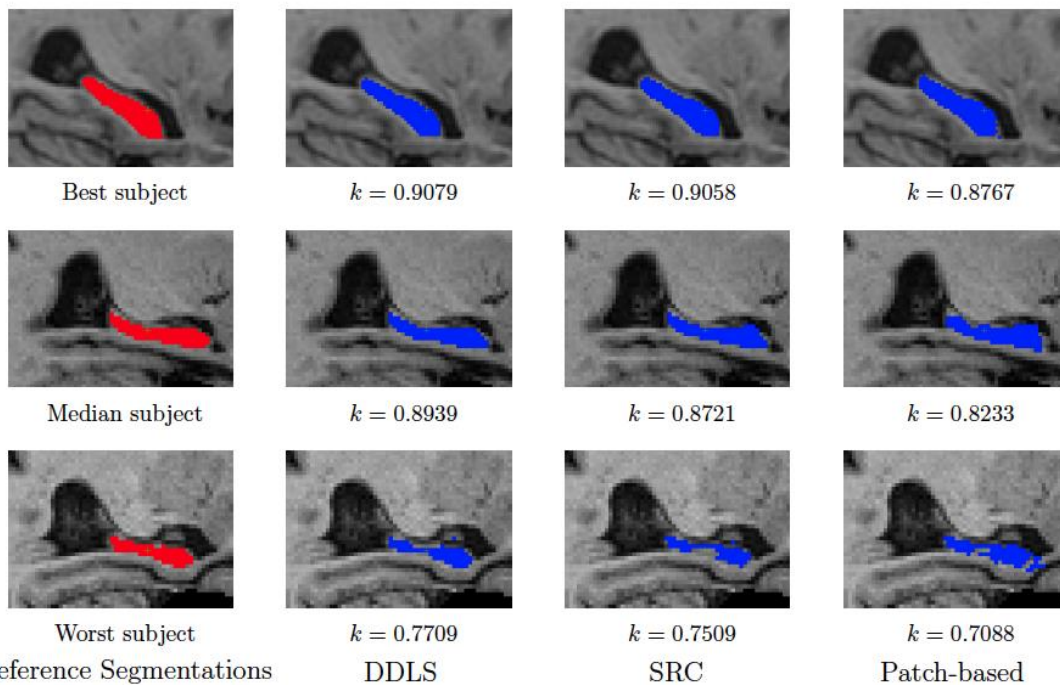


$k = 0.7088$

Automatic segmentations

Patch-based Labeling vs. Sparse Representation Classification (SRC) vs. Discriminative Dictionary Learning for Segmentation (DDLs)

Comparison of methods on ADNI dataset using Dice



Medical Image Analysis

(Multi)Atlas-Patch based Segmentation

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Medical Image Analysis Group